

# CALL

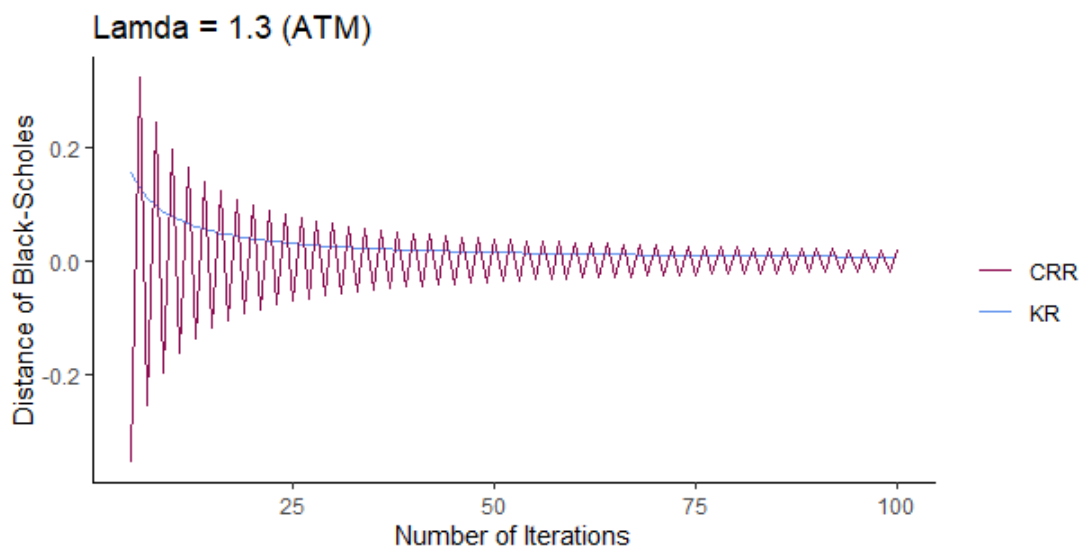
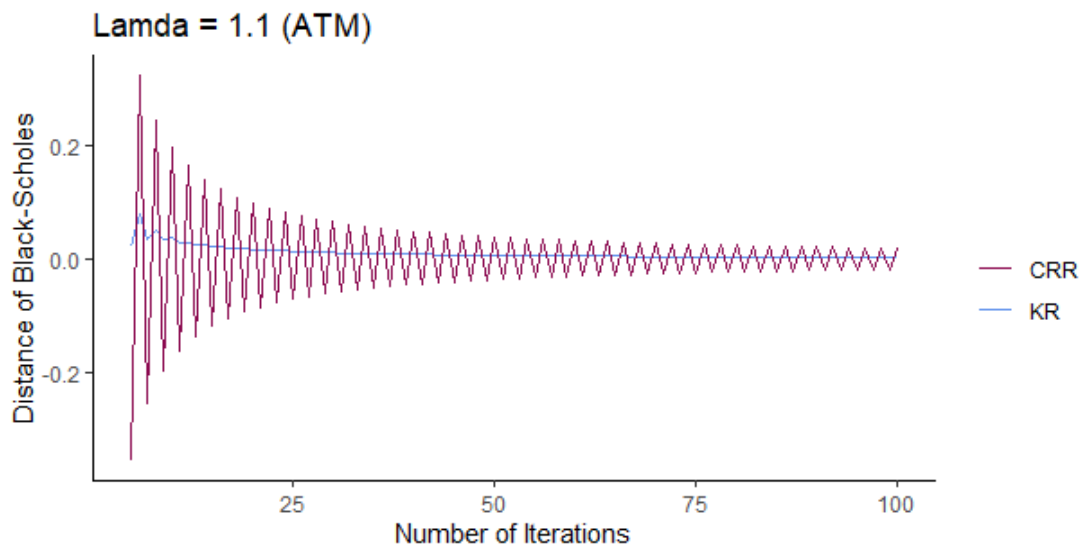
## MAIN MATRIX

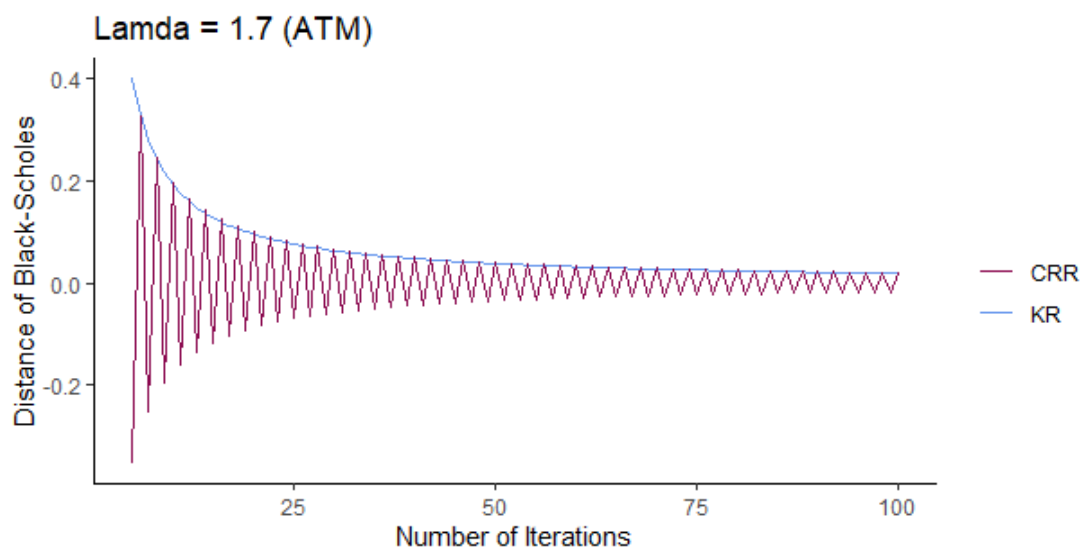
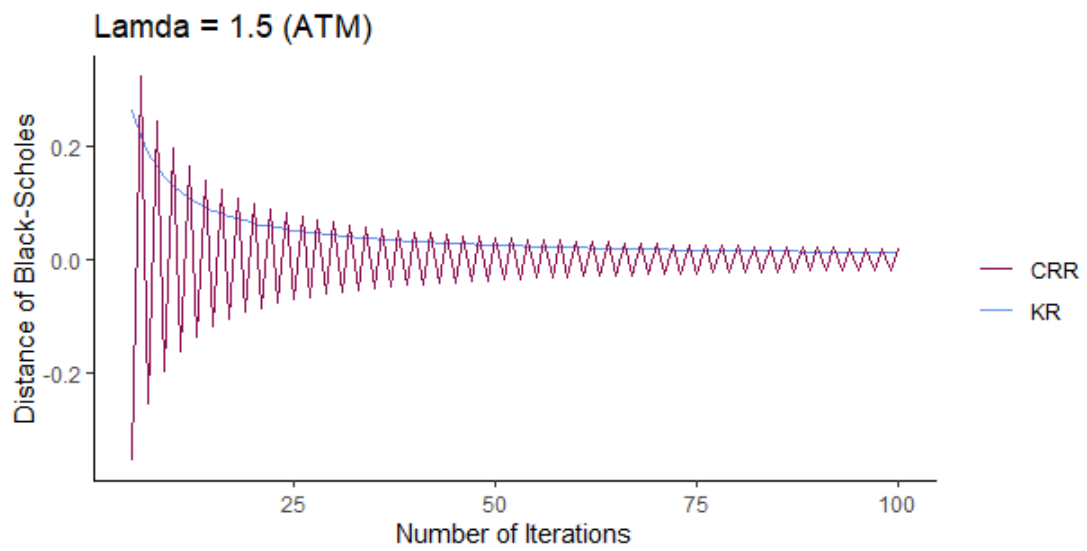
| $S_0 = 100$ $\text{Time} = 1$ $\text{Sigma} = 0.02$ $\text{Lamda} = 1.22474$ $\text{Risk free} = 0.05$ |         |         |         |         |
|--------------------------------------------------------------------------------------------------------|---------|---------|---------|---------|
| Number of iterations                                                                                   | CRR     | KR      | DiffCRR | DiffKR  |
| <b>K = 90</b> <b>BS = 16,6994</b>                                                                      |         |         |         |         |
| 5                                                                                                      | 16,5590 | 16,6374 | 0,1404  | 0,0621  |
| 10                                                                                                     | 16,7222 | 16,7419 | -0,0227 | -0,0425 |
| 20                                                                                                     | 16,7006 | 16,6901 | -0,0012 | 0,0093  |
| 40                                                                                                     | 16,7252 | 16,7081 | -0,0258 | -0,0086 |
| 60                                                                                                     | 16,6818 | 16,7061 | 0,0176  | -0,0066 |
| 80                                                                                                     | 16,7119 | 16,7000 | -0,0125 | -0,0006 |
| 100                                                                                                    | 16,7105 | 16,7030 | -0,0110 | -0,0035 |
| <b>K = 100</b> <b>BS = 10,4506</b>                                                                     |         |         |         |         |
| 5                                                                                                      | 10,8059 | 10,3320 | -0,3554 | 0,1185  |
| 10                                                                                                     | 10,2534 | 10,3887 | 0,1972  | 0,0618  |
| 20                                                                                                     | 10,3513 | 10,4192 | 0,0993  | 0,0314  |
| 40                                                                                                     | 10,4008 | 10,4348 | 0,0498  | 0,0158  |
| 60                                                                                                     | 10,4173 | 10,4400 | 0,0333  | 0,0106  |
| 80                                                                                                     | 10,4256 | 10,4427 | 0,0250  | 0,0079  |
| 100                                                                                                    | 10,4306 | 10,4442 | 0,0200  | 0,0064  |
| <b>K = 110</b> <b>BS = 6,0401</b>                                                                      |         |         |         |         |
| 5                                                                                                      | 5,6518  | 6,0020  | 0,3883  | 0,0380  |
| 10                                                                                                     | 6,0992  | 6,0562  | -0,0591 | -0,0161 |
| 20                                                                                                     | 5,9662  | 6,0537  | 0,0739  | -0,0136 |
| 40                                                                                                     | 6,0789  | 6,0549  | -0,0388 | -0,0148 |
| 60                                                                                                     | 6,0347  | 6,0267  | 0,0054  | 0,0133  |

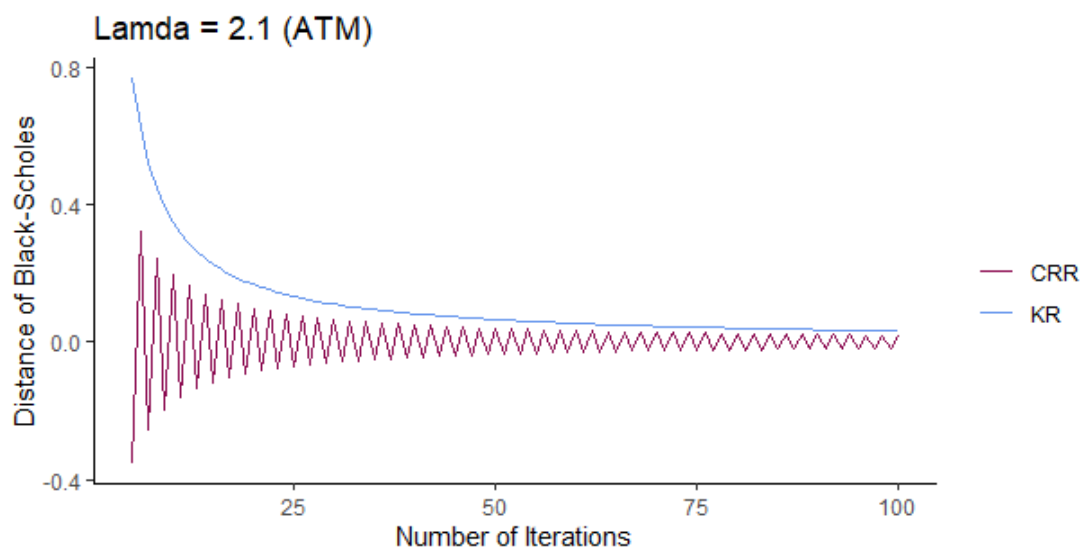
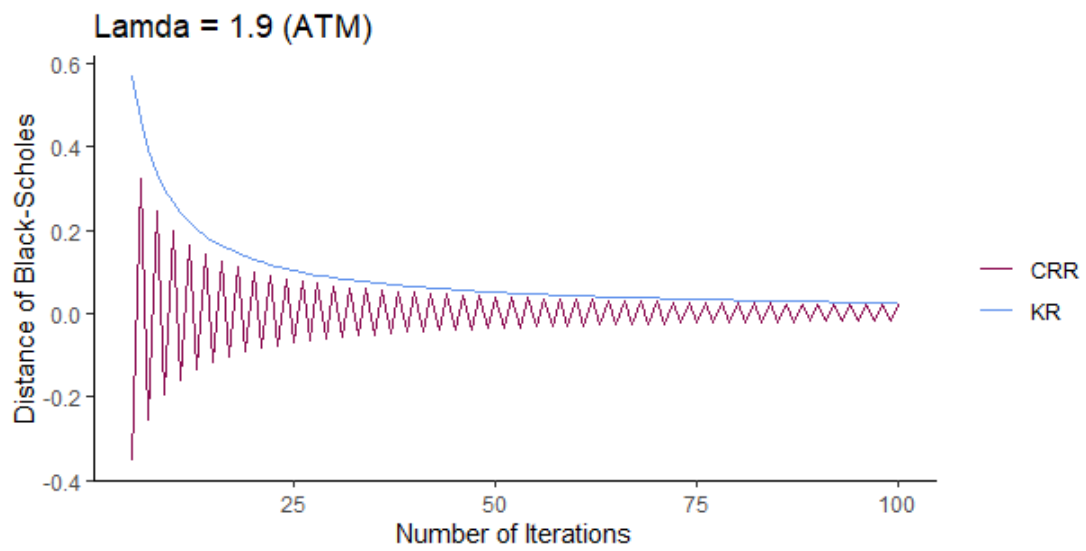
|     |        |        |         |         |
|-----|--------|--------|---------|---------|
| 80  | 6,0322 | 6,0476 | 0,0079  | -0,0075 |
| 100 | 6,0532 | 6,0371 | -0,0131 | 0,0030  |

## PLOTS (DIFFERENT LAMDA)

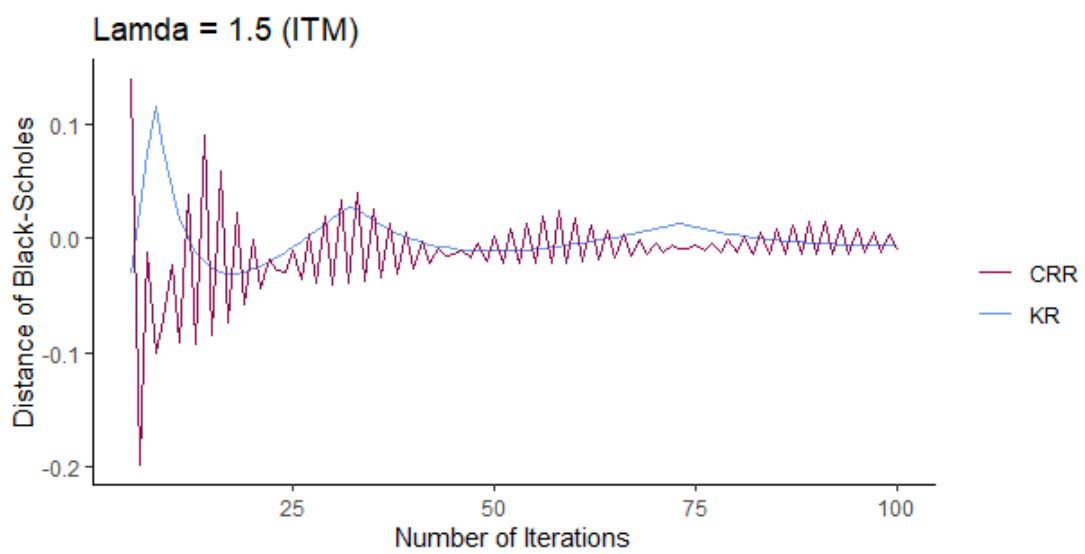
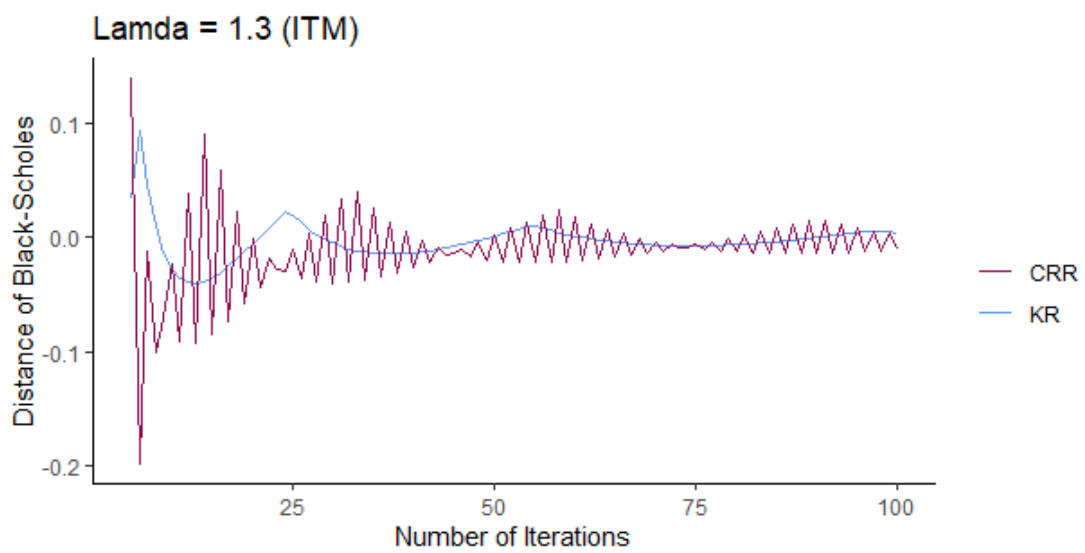
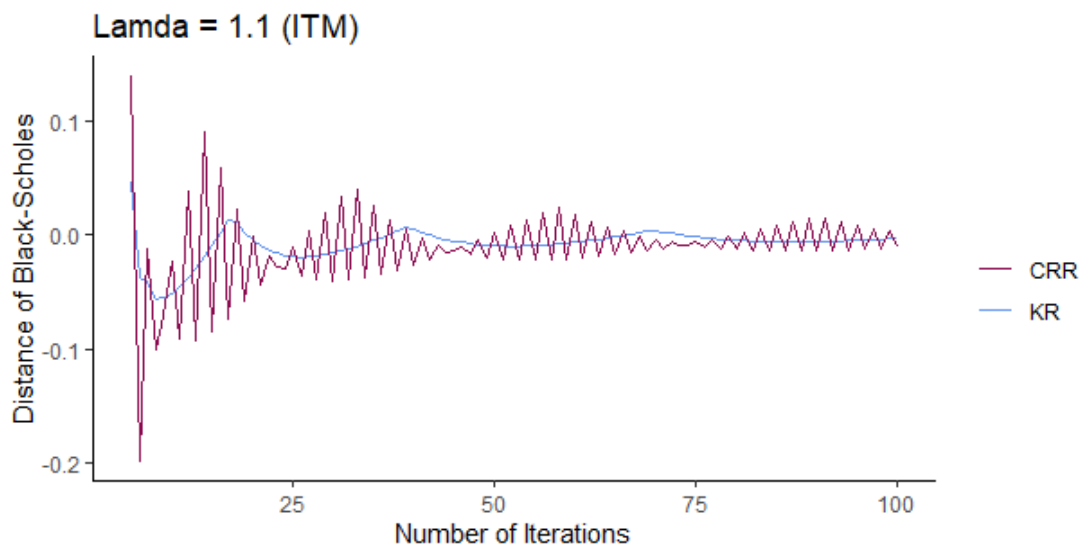
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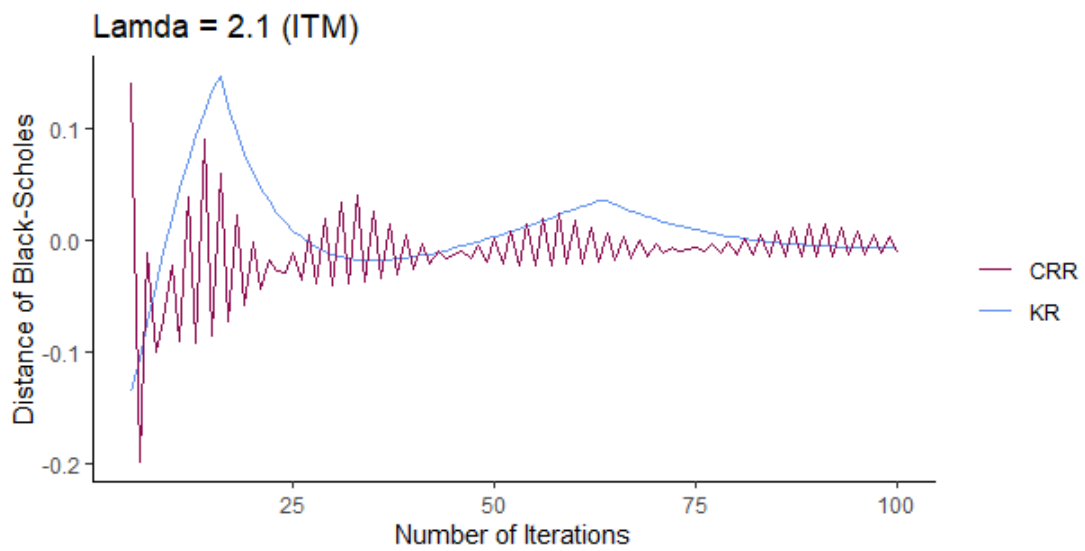
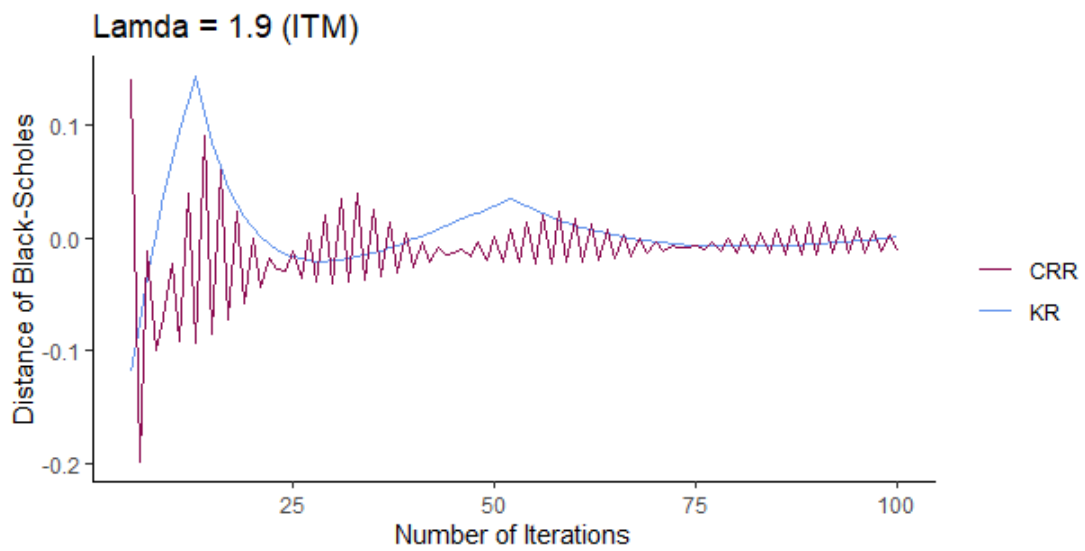
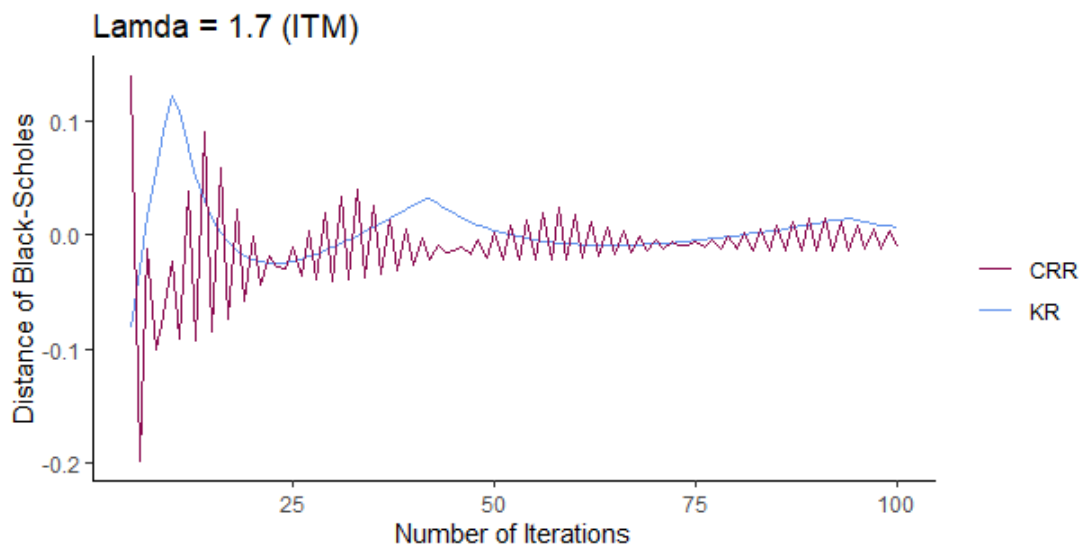




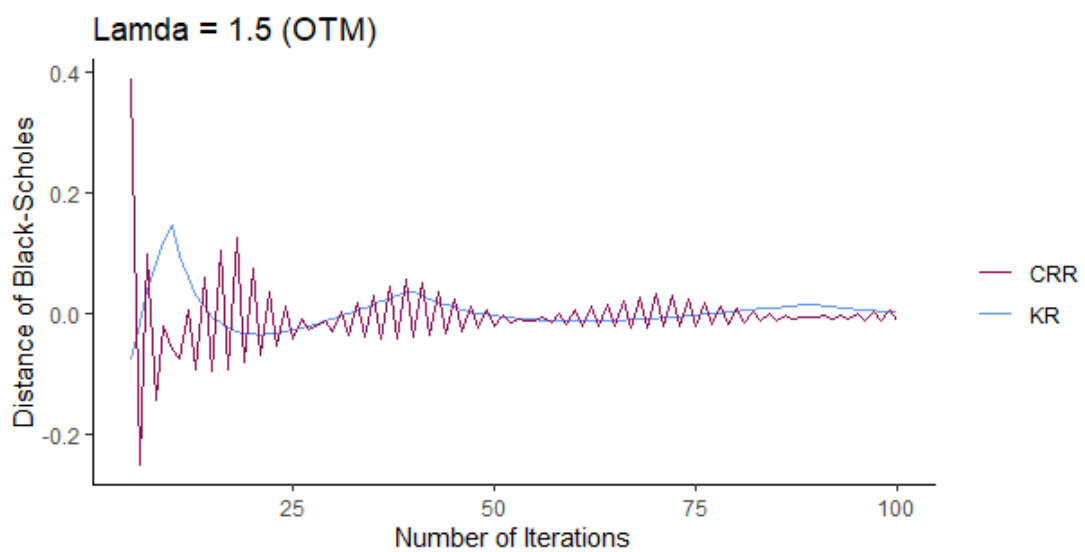
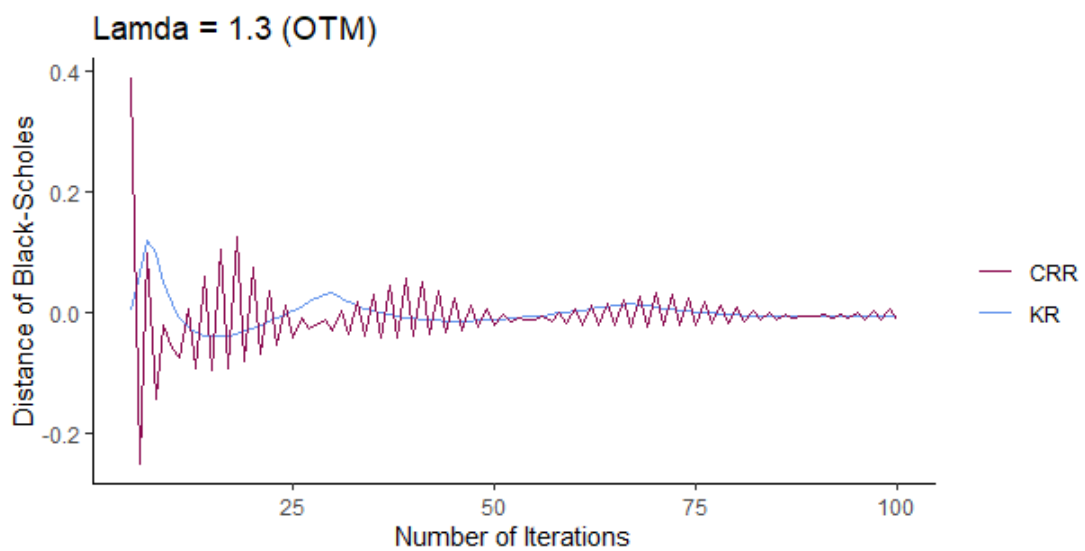
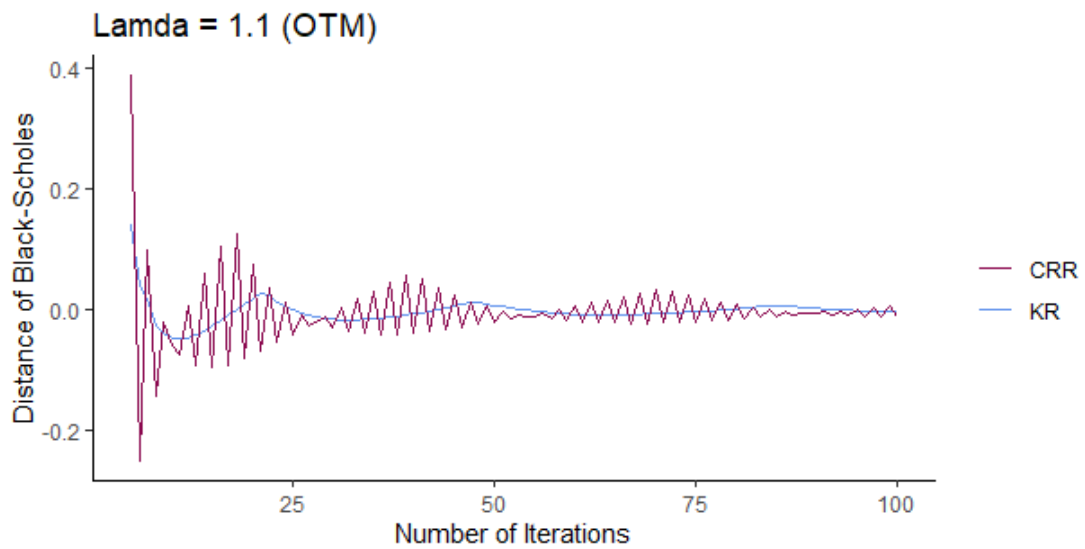


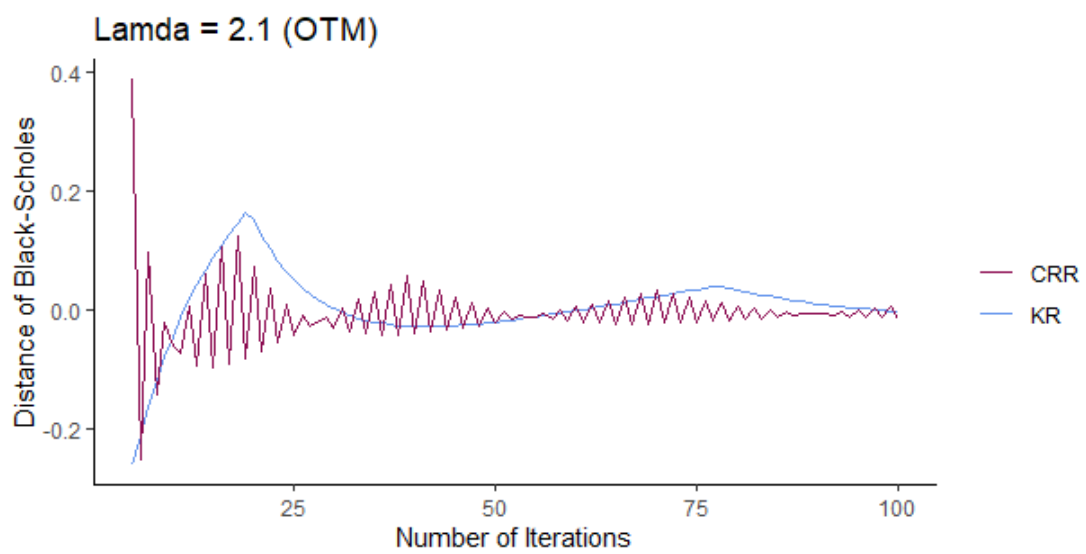
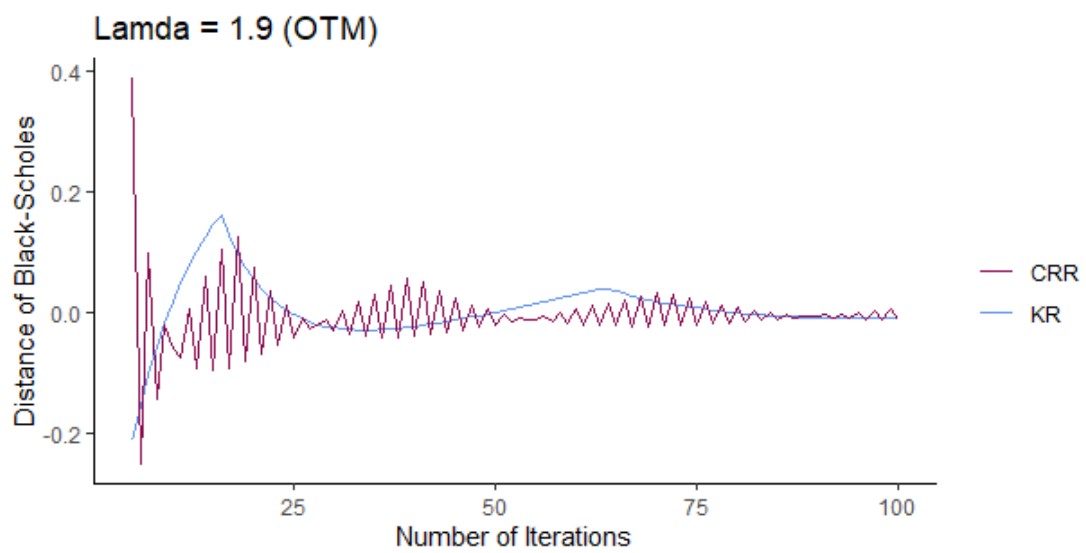
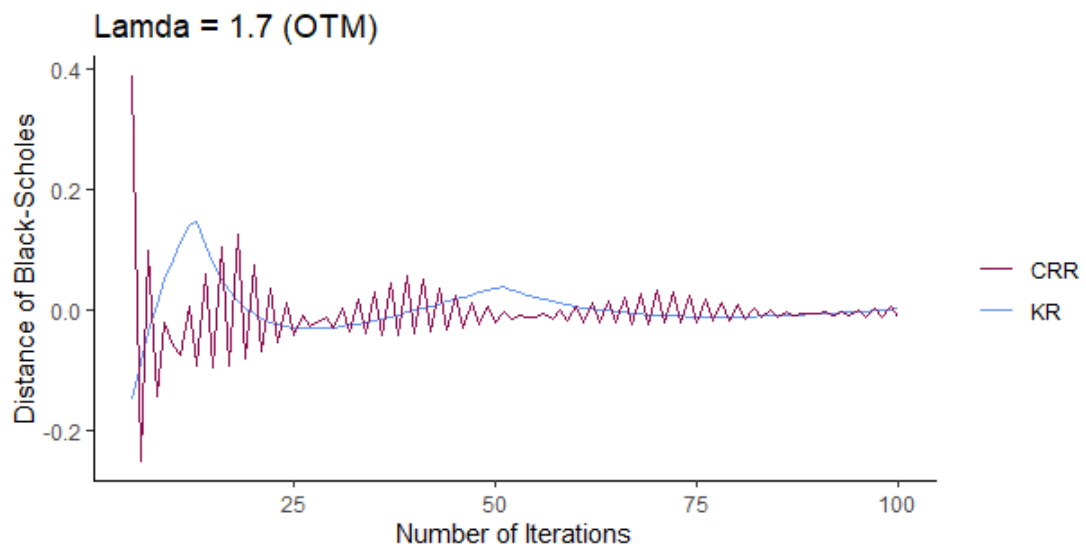
**IN THE MONEY**





## OUT THE MONEY



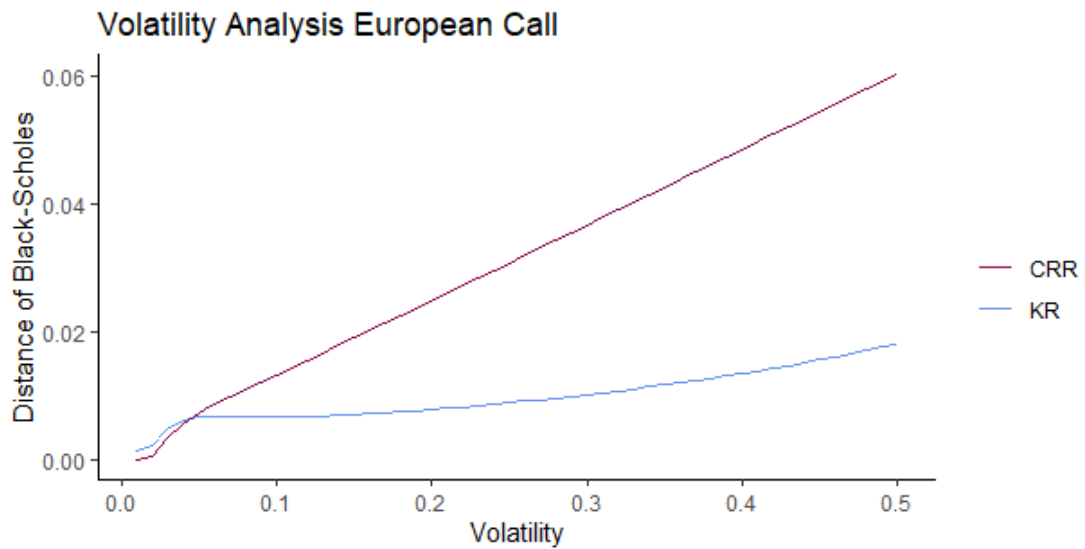




## VOLATILITY ANALYSIS

| $S_0 = 100$ $K = 100$ Risk Free = 0.05      Lamda = 1.22474      Time = 1 |         |         |         |        |
|---------------------------------------------------------------------------|---------|---------|---------|--------|
| Number of iterations                                                      | CRR     | KR      | DiffCRR | DiffKR |
| <b>Sigma = 0.15      BS = 8,5917</b>                                      |         |         |         |        |
| 5                                                                         | 8,8380  | 8,4831  | -0,2464 | 0,1086 |
| 10                                                                        | 8,4406  | 8,5356  | 0,1510  | 0,0560 |
| 20                                                                        | 8,5156  | 8,5633  | 0,0761  | 0,0283 |
| 40                                                                        | 8,5535  | 8,5774  | 0,0382  | 0,0142 |
| 60                                                                        | 8,5662  | 8,5821  | 0,0255  | 0,0095 |
| 80                                                                        | 8,5725  | 8,5845  | 0,0191  | 0,0071 |
| 100                                                                       | 8,5764  | 8,5859  | 0,0153  | 0,0057 |
| <b>Sigma = 0.18      BS = 9,7023</b>                                      |         |         |         |        |
| 5                                                                         | 10,0149 | 9,5884  | -0,3126 | 0,1139 |
| 10                                                                        | 9,5237  | 9,6431  | 0,1786  | 0,0592 |
| 20                                                                        | 9,6123  | 9,6723  | 0,0900  | 0,0300 |
| 40                                                                        | 9,6572  | 9,6872  | 0,0452  | 0,0151 |
| 60                                                                        | 9,6722  | 9,6922  | 0,0301  | 0,0101 |
| 80                                                                        | 9,6797  | 9,6948  | 0,0226  | 0,0076 |
| 100                                                                       | 9,6842  | 9,6963  | 0,0181  | 0,0061 |
| <b>Sigma = 0.24      BS = 11,9578</b>                                     |         |         |         |        |
| 5                                                                         | 12,3961 | 11,8279 | -0,4383 | 0,1299 |
| 10                                                                        | 11,7234 | 11,8897 | 0,2344  | 0,0681 |
| 20                                                                        | 11,8397 | 11,9231 | 0,1181  | 0,0346 |
| 40                                                                        | 11,8985 | 11,9403 | 0,0592  | 0,0175 |
| 60                                                                        | 11,9182 | 11,9461 | 0,0395  | 0,0117 |

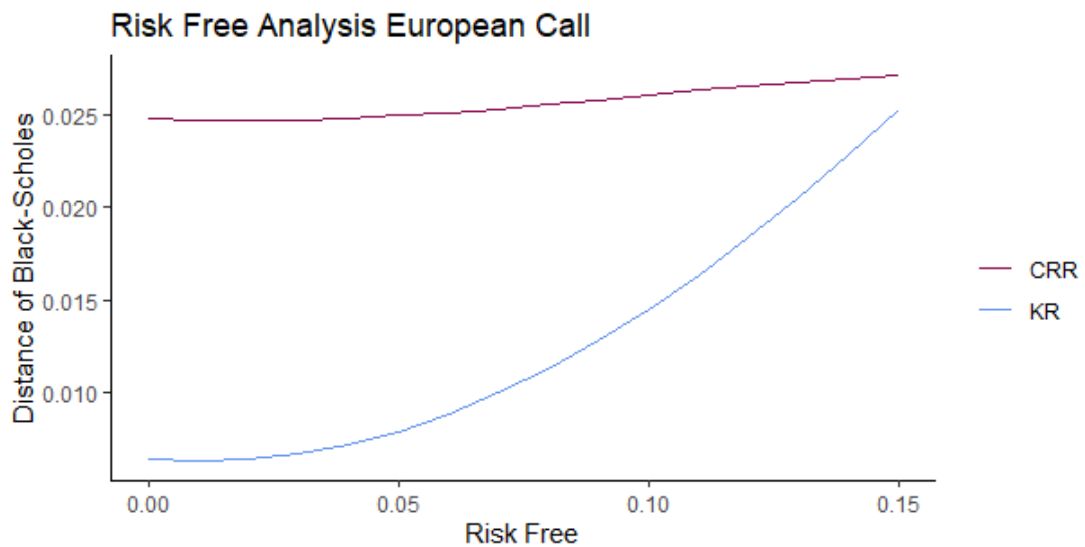
|     |         |         |        |        |
|-----|---------|---------|--------|--------|
| 80  | 11,9281 | 11,9490 | 0,0297 | 0,0088 |
| 100 | 11,9340 | 11,9508 | 0,0237 | 0,0070 |



## RISK FREE ANALYSIS

| $S_0 = 100$ $K = 100$ $\text{Sigma} = 0.02$ $\text{Lamda} = 1.22474$ $\text{Time} = 1$ |        |        |             |        |
|----------------------------------------------------------------------------------------|--------|--------|-------------|--------|
| Number of iterations                                                                   | CRR    | KR     | DiffCRR     | DiffKR |
| Risk free = 0.01                                                                       |        |        | BS = 8,4333 |        |
| 5                                                                                      | 8,8327 | 8,3416 | -0,3994     | 0,0917 |
| 10                                                                                     | 8,2377 | 8,3847 | 0,1956      | 0,0487 |
| 20                                                                                     | 8,3348 | 8,4085 | 0,0985      | 0,0248 |
| 40                                                                                     | 8,3839 | 8,4208 | 0,0494      | 0,0125 |
| 60                                                                                     | 8,4003 | 8,4249 | 0,0330      | 0,0084 |
| 80                                                                                     | 8,4086 | 8,4270 | 0,0248      | 0,0063 |
| 100                                                                                    | 8,4135 | 8,4283 | 0,0198      | 0,0050 |

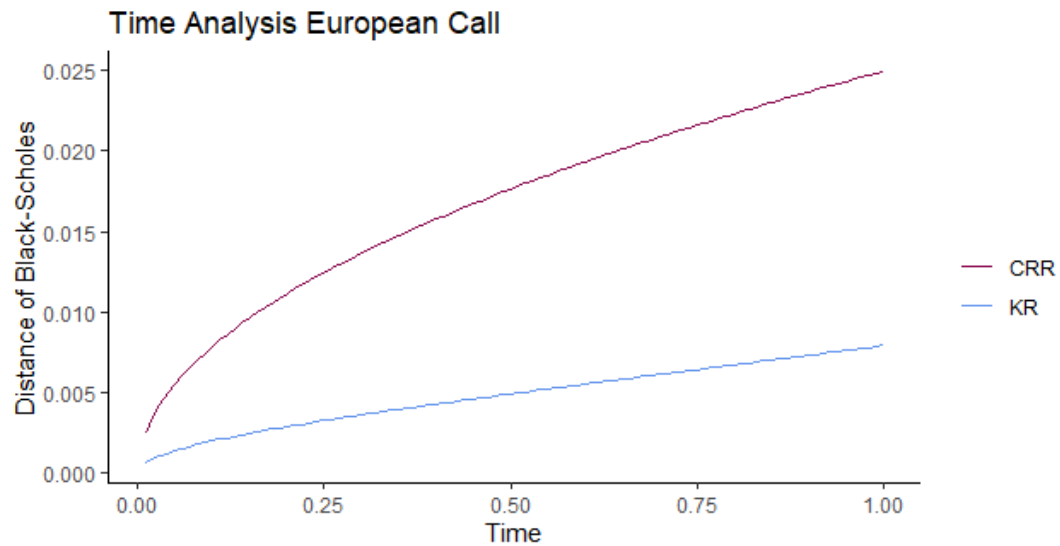
| Risk free = 0.04      BS = 9,9251  |         |         |         |        |
|------------------------------------|---------|---------|---------|--------|
| 5                                  | 10,2956 | 9,8184  | -0,3706 | 0,1066 |
| 10                                 | 9,7289  | 9,8691  | 0,1962  | 0,0560 |
| 20                                 | 9,8262  | 9,8966  | 0,0988  | 0,0285 |
| 40                                 | 9,8755  | 9,9107  | 0,0496  | 0,0144 |
| 60                                 | 9,8920  | 9,9155  | 0,0331  | 0,0096 |
| 80                                 | 9,9002  | 9,9178  | 0,0248  | 0,0072 |
| 100                                | 9,9052  | 9,9193  | 0,0199  | 0,0058 |
| Risk free = 0.07      BS = 11,5415 |         |         |         |        |
| 5                                  | 11,8590 | 11,3891 | -0,3175 | 0,1524 |
| 10                                 | 11,3414 | 11,4630 | 0,2001  | 0,0785 |
| 20                                 | 11,4407 | 11,5018 | 0,1008  | 0,0397 |
| 40                                 | 11,4909 | 11,5215 | 0,0506  | 0,0199 |
| 60                                 | 11,5077 | 11,5282 | 0,0338  | 0,0133 |
| 80                                 | 11,5161 | 11,5315 | 0,0253  | 0,0100 |
| 100                                | 11,5212 | 11,5335 | 0,0203  | 0,0080 |



## TIME ANALYSIS

| $S_0 = 100$ $K = 100$ $\text{Sigma} = 0.02$ $\text{Lamda} = 1.22474$ $\text{Risk free} = 0.05$ |        |        |         |        |
|------------------------------------------------------------------------------------------------|--------|--------|---------|--------|
| Number of iterations                                                                           | CRR    | KR     | DiffCRR | DiffKR |
| <b>Time = 2/12      BS = 3,6747</b>                                                            |        |        |         |        |
| 5                                                                                              | 3,8370 | 3,6365 | -0,1623 | 0,0382 |
| 10                                                                                             | 3,5944 | 3,6545 | 0,0804  | 0,0203 |
| 20                                                                                             | 3,6343 | 3,6644 | 0,0405  | 0,0103 |
| 40                                                                                             | 3,6544 | 3,6695 | 0,0203  | 0,0052 |
| 60                                                                                             | 3,6612 | 3,6712 | 0,0136  | 0,0035 |
| 80                                                                                             | 3,6646 | 3,6721 | 0,0102  | 0,0026 |
| 100                                                                                            | 3,6666 | 3,6726 | 0,0081  | 0,0021 |
| <b>Time = 5/12      BS = 6,1907</b>                                                            |        |        |         |        |
| 5                                                                                              | 6,4389 | 6,1259 | -0,2482 | 0,0648 |
| 10                                                                                             | 6,0635 | 6,1565 | 0,1272  | 0,0341 |
| 20                                                                                             | 6,1266 | 6,1733 | 0,0641  | 0,0174 |
| 40                                                                                             | 6,1585 | 6,1819 | 0,0321  | 0,0088 |
| 60                                                                                             | 6,1692 | 6,1848 | 0,0214  | 0,0059 |
| 80                                                                                             | 6,1746 | 6,1863 | 0,0161  | 0,0044 |
| 100                                                                                            | 6,1778 | 6,1871 | 0,0129  | 0,0035 |
| <b>Time = 8/12      BS = 8,1726</b>                                                            |        |        |         |        |
| 5                                                                                              | 8,4762 | 8,0846 | -0,3036 | 0,0880 |
| 10                                                                                             | 8,0116 | 8,1264 | 0,1609  | 0,0462 |
| 20                                                                                             | 8,0915 | 8,1491 | 0,0811  | 0,0235 |
| 40                                                                                             | 8,1319 | 8,1607 | 0,0407  | 0,0118 |
| 60                                                                                             | 8,1454 | 8,1647 | 0,0271  | 0,0079 |

|     |        |        |        |        |
|-----|--------|--------|--------|--------|
| 80  | 8,1522 | 8,1666 | 0,0204 | 0,0059 |
| 100 | 8,1563 | 8,1678 | 0,0163 | 0,0048 |



## EUROPEAN PUT

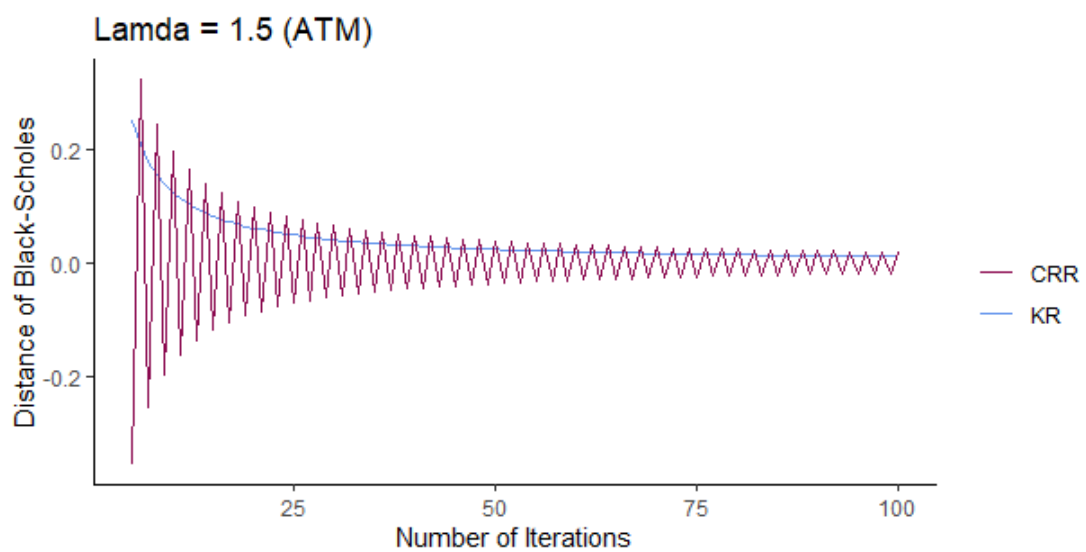
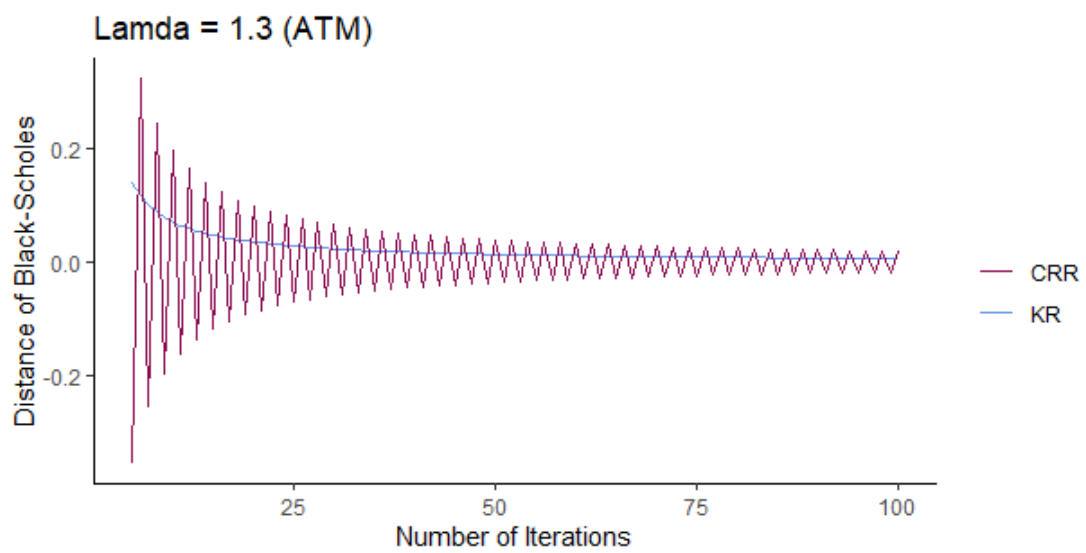
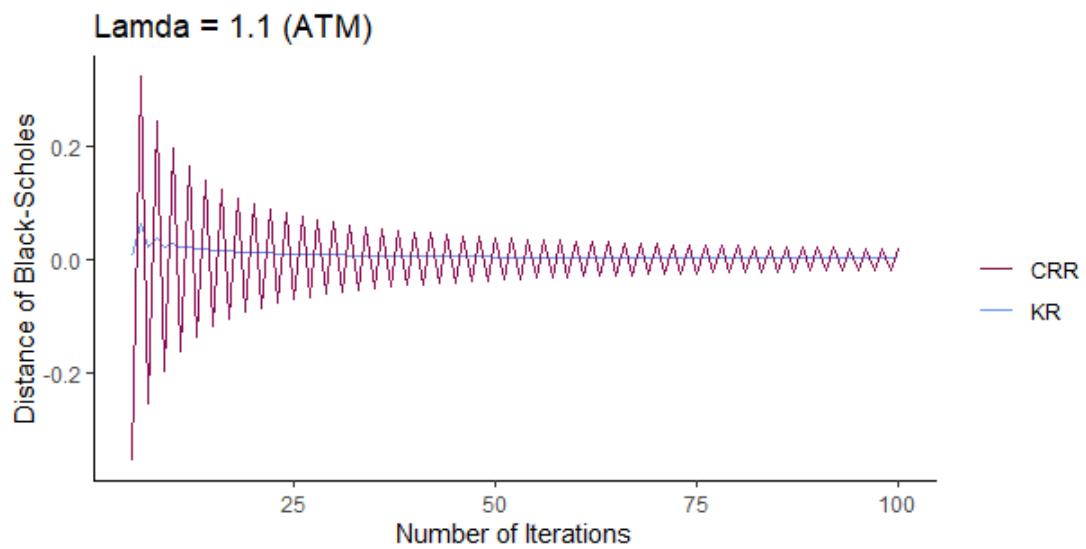
### MAIN MATRIX

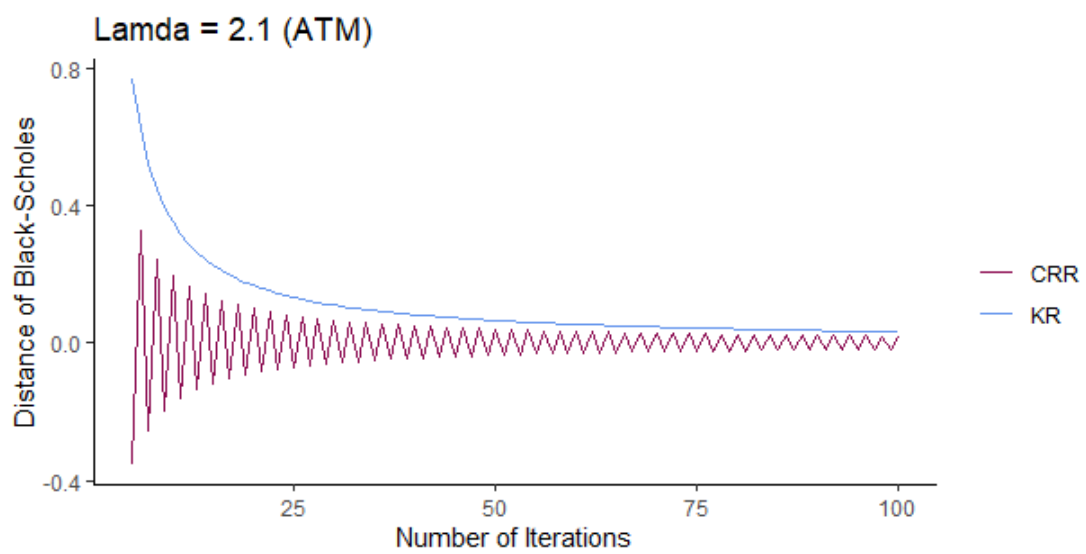
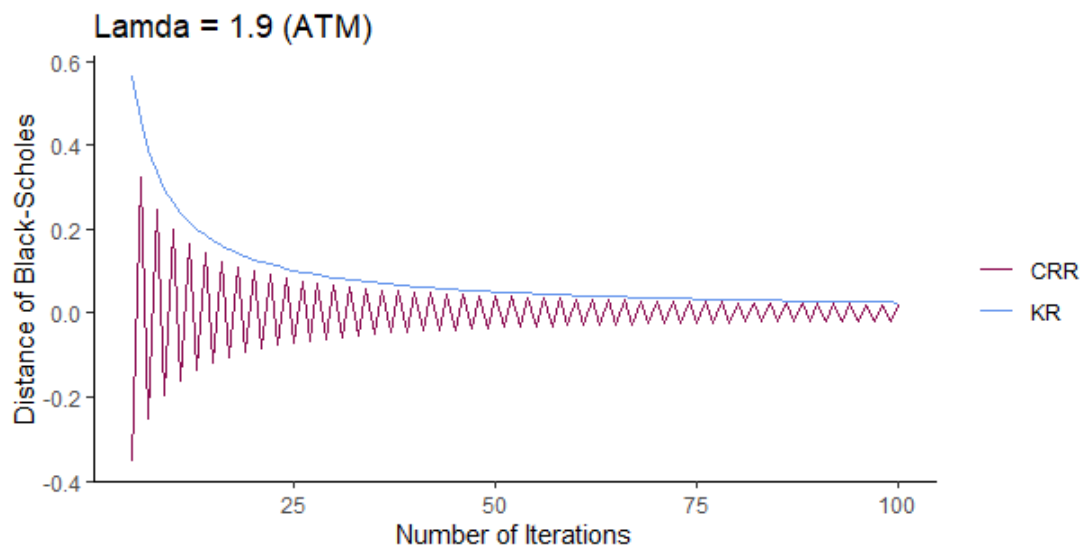
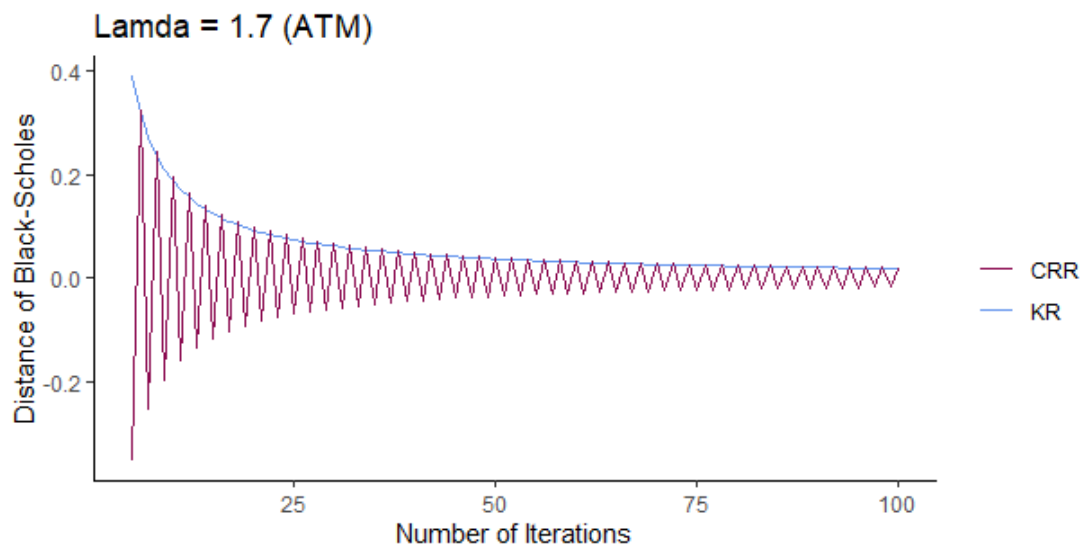
|                      |          |              |                 |                  |
|----------------------|----------|--------------|-----------------|------------------|
| $S_0 = 100$          | Time = 1 | Sigma = 0.02 | Lamda = 1.22474 | Risk free = 0.05 |
| Number of iterations | CRR      | KR           | DiffCRR         | DiffKR           |

| K = 90 BS = 2,3101   |         |         |         |         |
|----------------------|---------|---------|---------|---------|
| 5                    | 2,1697  | 2,2649  | 0,1404  | 0,0452  |
| 10                   | 2,3328  | 2,3610  | -0,0227 | -0,0509 |
| 20                   | 2,3113  | 2,3050  | -0,0012 | 0,0051  |
| 40                   | 2,3359  | 2,3208  | -0,0258 | -0,0107 |
| 60                   | 2,2925  | 2,3181  | 0,0176  | -0,0080 |
| 80                   | 2,3226  | 2,3117  | -0,0125 | -0,0016 |
| 100                  | 2,3211  | 2,3145  | -0,0110 | -0,0044 |
| K = 100 BS = 5,5735  |         |         |         |         |
| 5                    | 5,9289  | 5,4719  | -0,3554 | 0,1016  |
| 10                   | 5,3764  | 5,5202  | 0,1972  | 0,0534  |
| 20                   | 5,4742  | 5,5464  | 0,0993  | 0,0271  |
| 40                   | 5,5237  | 5,5598  | 0,0498  | 0,0137  |
| 60                   | 5,5403  | 5,5644  | 0,0333  | 0,0091  |
| 80                   | 5,5486  | 5,5667  | 0,0250  | 0,0069  |
| 100                  | 5,5536  | 5,5680  | 0,0200  | 0,0055  |
| K = 110 BS = 10,6753 |         |         |         |         |
| 5                    | 10,2871 | 10,6542 | 0,3883  | 0,0211  |
| 10                   | 10,7344 | 10,6999 | -0,0591 | -0,0245 |
| 20                   | 10,6015 | 10,6932 | 0,0739  | -0,0178 |
| 40                   | 10,7141 | 10,6923 | -0,0388 | -0,0169 |
| 60                   | 10,6699 | 10,6634 | 0,0054  | 0,0119  |
| 80                   | 10,6674 | 10,6839 | 0,0079  | -0,0085 |
| 100                  | 10,6885 | 10,6732 | -0,0131 | 0,0022  |

**PLOTS (DIFFERENT LAMDA)**

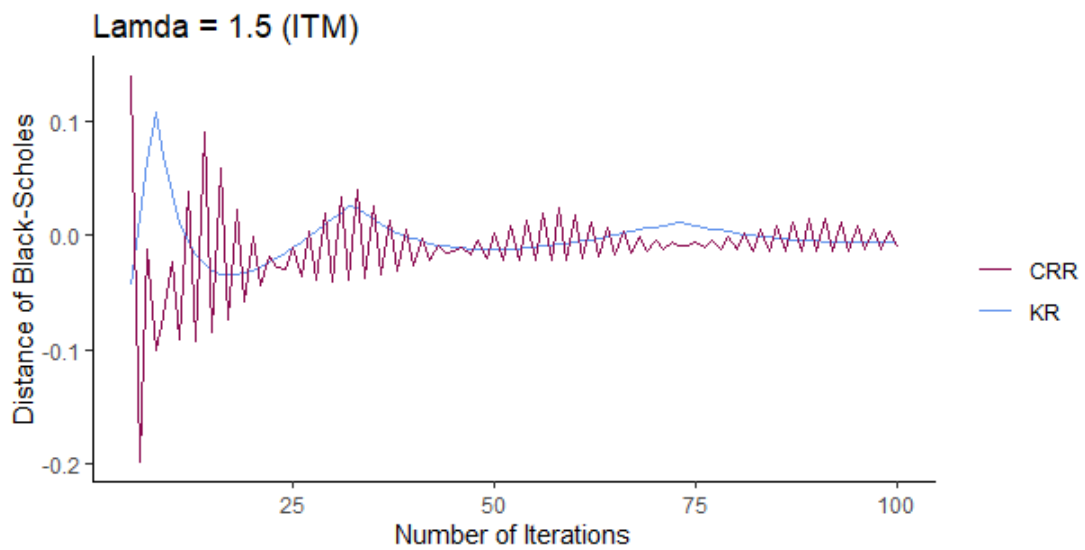
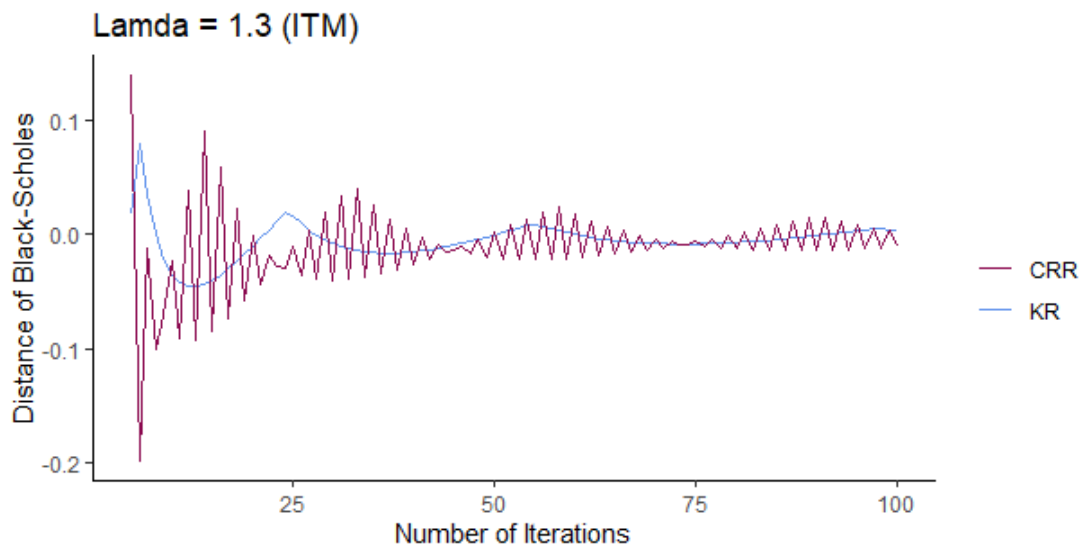
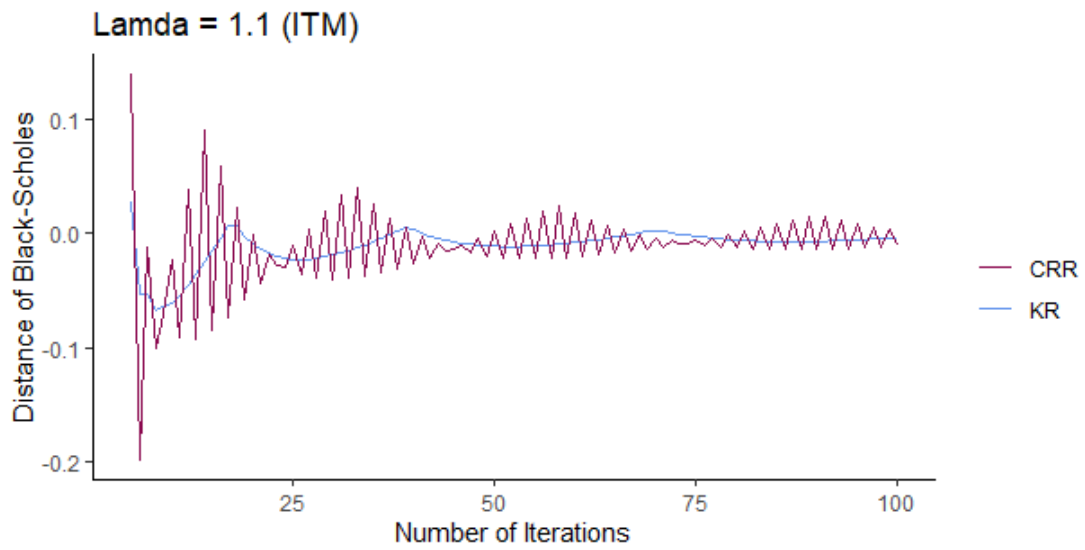
**AT THE MONEY**

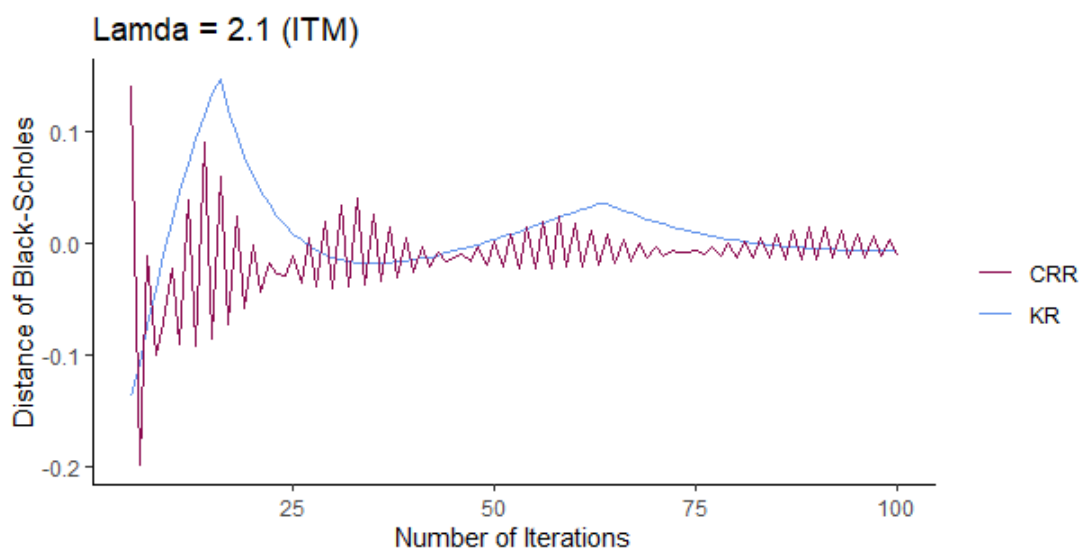
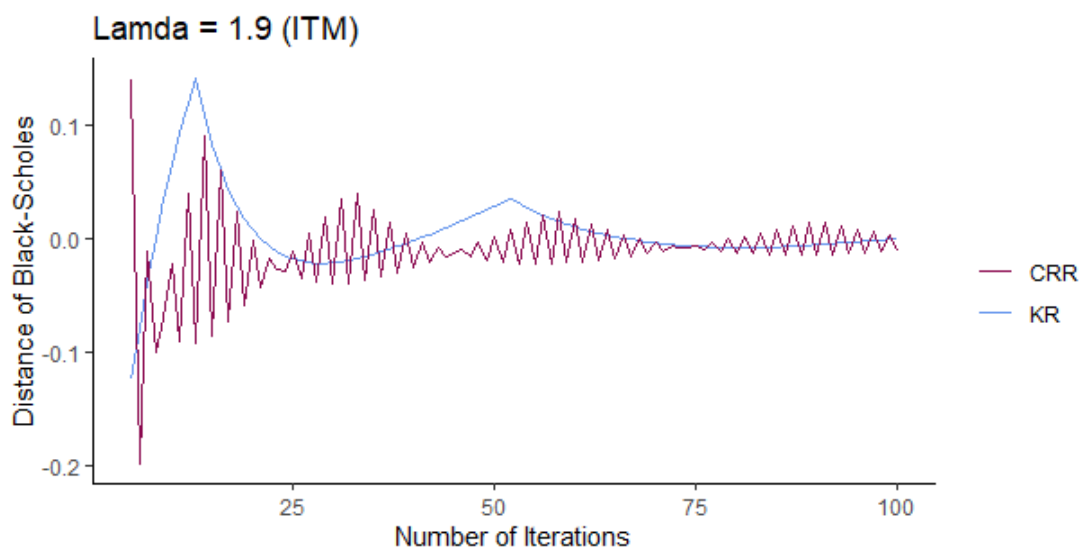
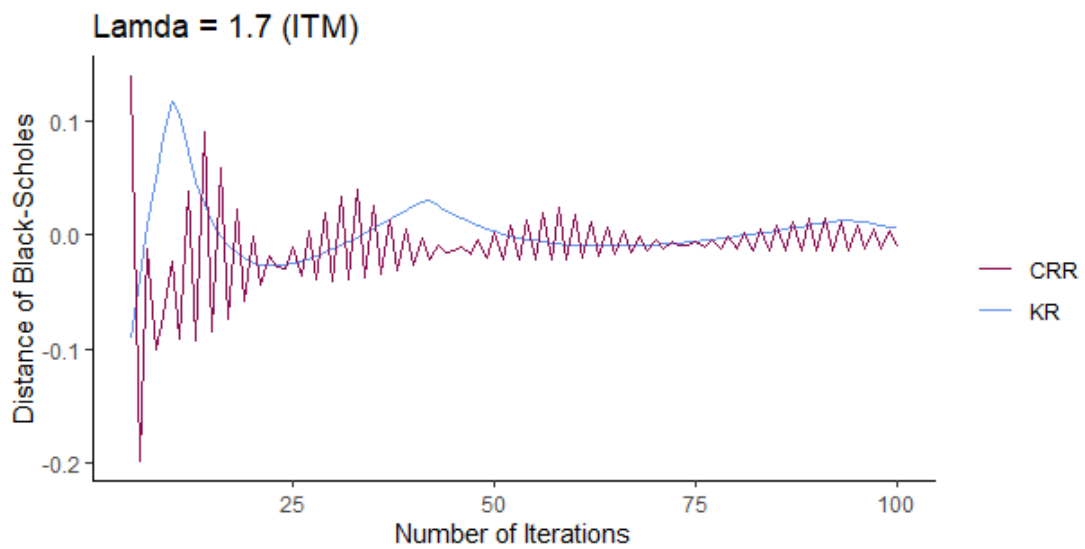




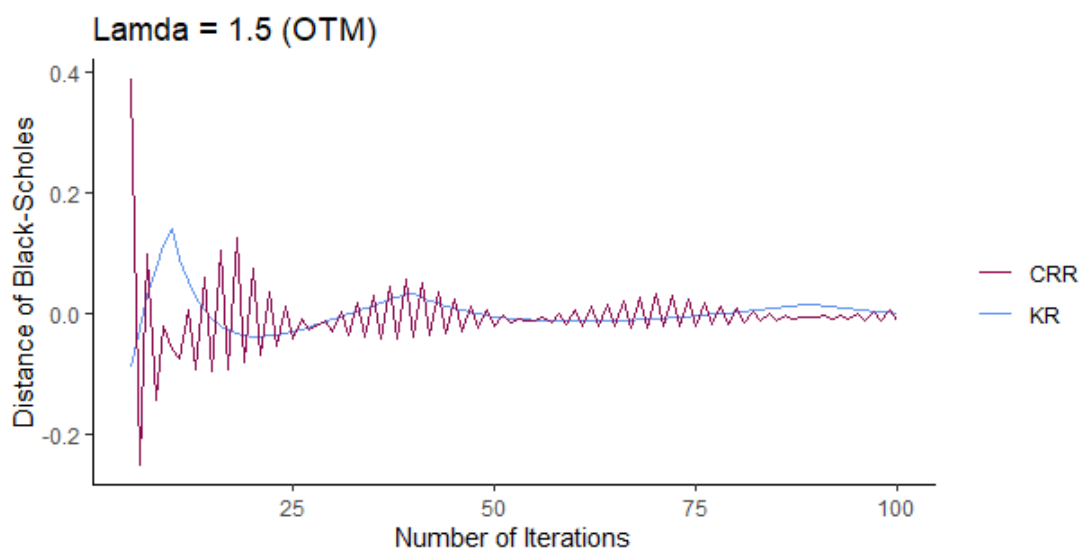
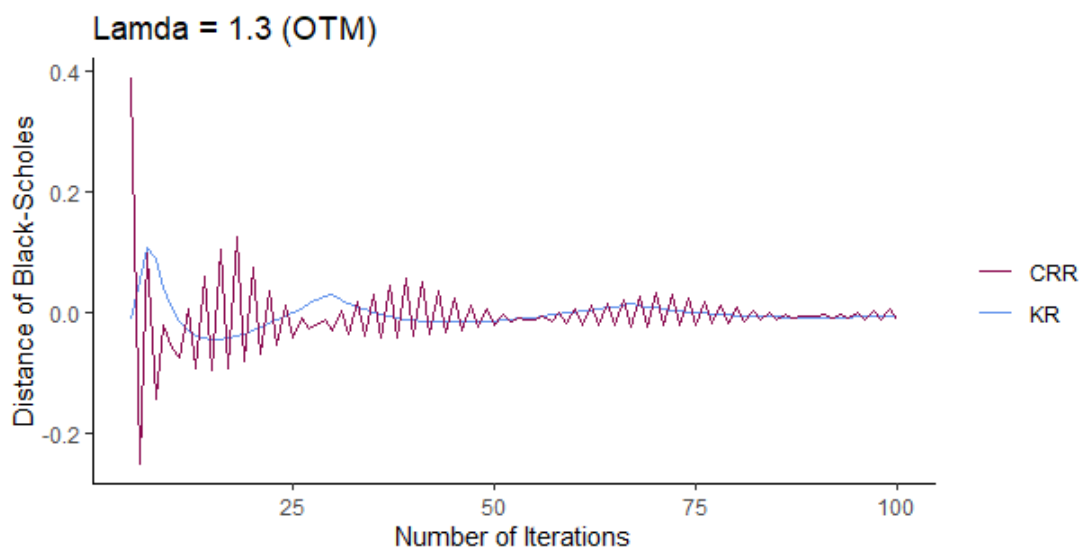
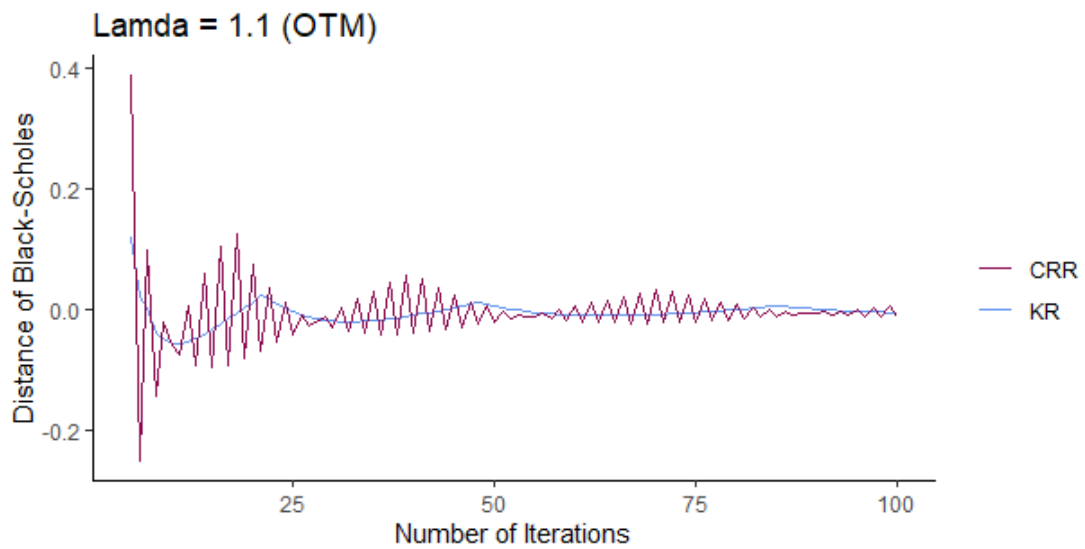


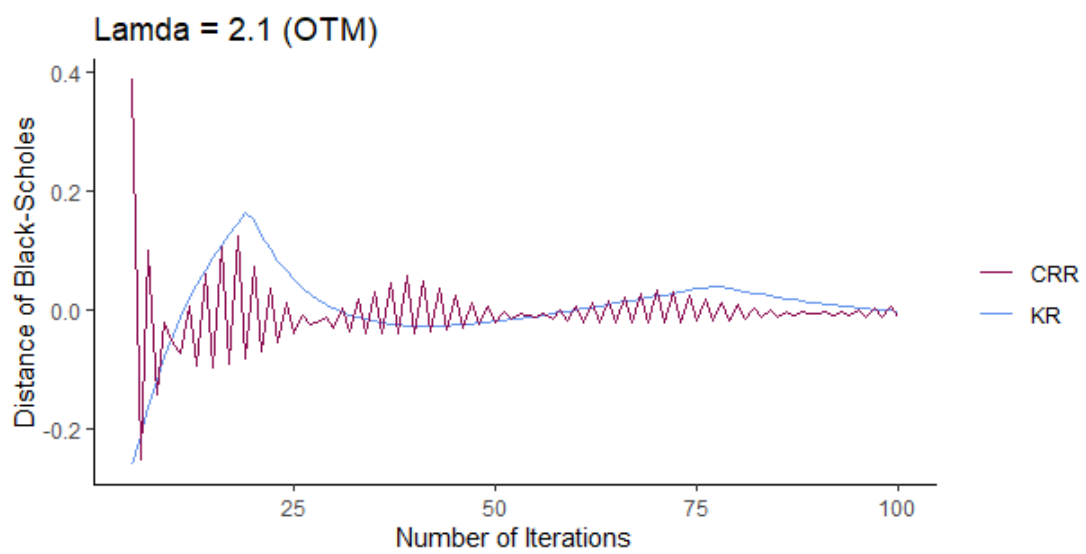
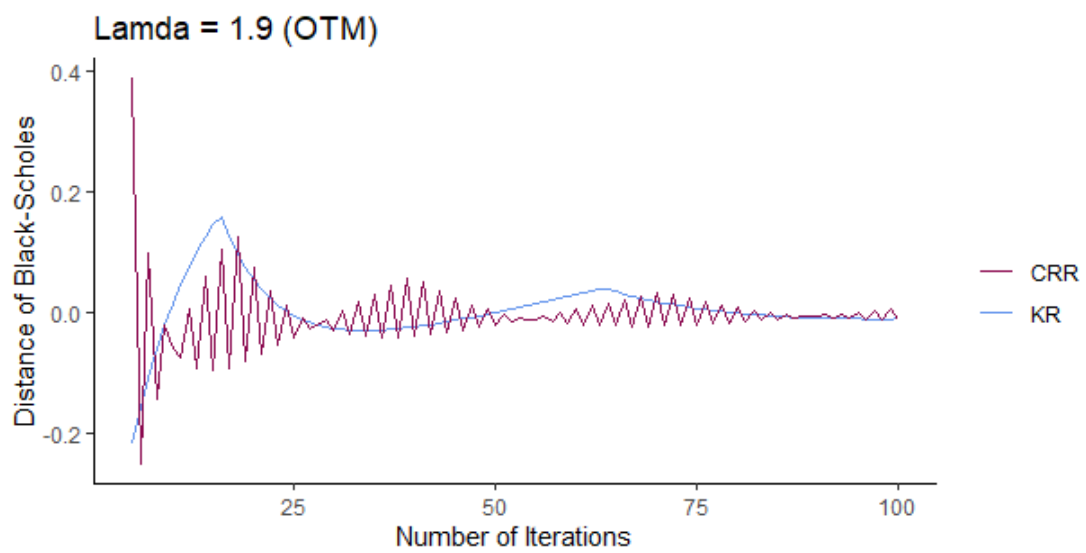
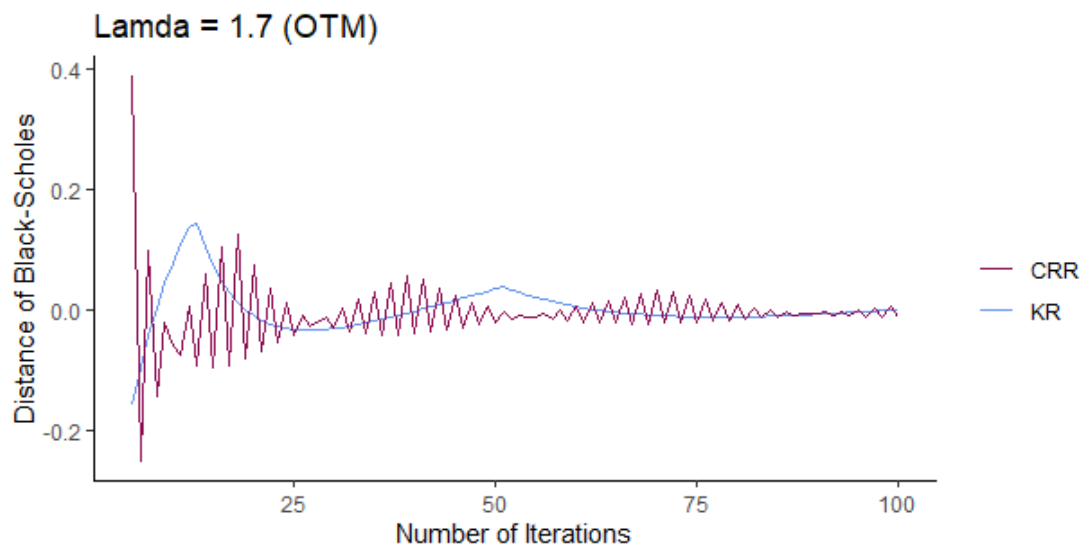
## IN THE MONEY





## OUT THE MONEY

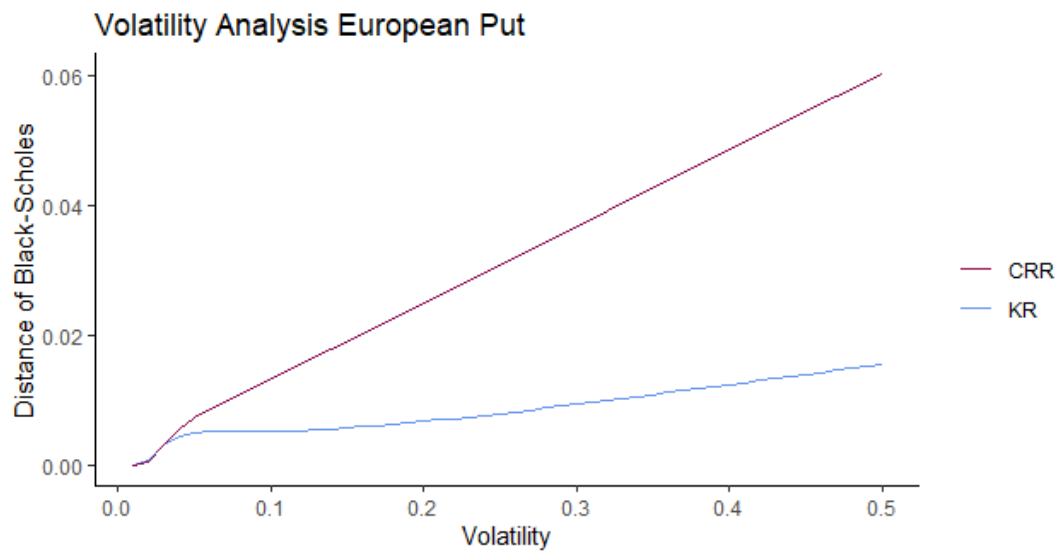




## VOLATILITY ANALYSIS

| $S_0 = 100$ $K = 100$ Risk Free = 0.05      Lamda = 1.22474      Time = 1 |        |        |         |        |
|---------------------------------------------------------------------------|--------|--------|---------|--------|
| Number of iterations                                                      | CRR    | KR     | DiffCRR | DiffKR |
| <b>Sigma = 0.15      BS = 3,7146</b>                                      |        |        |         |        |
| 5                                                                         | 3,9610 | 3,6259 | -0,2464 | 0,0887 |
| 10                                                                        | 3,5636 | 3,6686 | 0,1510  | 0,0460 |
| 20                                                                        | 3,6385 | 3,6913 | 0,0761  | 0,0233 |
| 40                                                                        | 3,6764 | 3,7029 | 0,0382  | 0,0117 |
| 60                                                                        | 3,6891 | 3,7068 | 0,0255  | 0,0078 |
| 80                                                                        | 3,6955 | 3,7087 | 0,0191  | 0,0059 |
| 100                                                                       | 3,6993 | 3,7099 | 0,0153  | 0,0047 |
| <b>Sigma = 0.18      BS = 4,8253</b>                                      |        |        |         |        |
| 5                                                                         | 5,1379 | 4,7295 | -0,3126 | 0,0958 |
| 10                                                                        | 4,6466 | 4,7752 | 0,1786  | 0,0501 |
| 20                                                                        | 4,7353 | 4,7998 | 0,0900  | 0,0255 |
| 40                                                                        | 4,7801 | 4,8125 | 0,0452  | 0,0128 |
| 60                                                                        | 4,7952 | 4,8167 | 0,0301  | 0,0086 |
| 80                                                                        | 4,8027 | 4,8188 | 0,0226  | 0,0064 |
| 100                                                                       | 4,8072 | 4,8201 | 0,0181  | 0,0052 |
| <b>Sigma = 0.24      BS = 7,0807</b>                                      |        |        |         |        |
| 5                                                                         | 7,5191 | 6,9655 | -0,4383 | 0,1152 |

|     |        |        |        |        |
|-----|--------|--------|--------|--------|
| 10  | 6,8463 | 7,0200 | 0,2344 | 0,0608 |
| 20  | 6,9626 | 7,0498 | 0,1181 | 0,0310 |
| 40  | 7,0215 | 7,0651 | 0,0592 | 0,0156 |
| 60  | 7,0412 | 7,0703 | 0,0395 | 0,0104 |
| 80  | 7,0511 | 7,0729 | 0,0297 | 0,0078 |
| 100 | 7,0570 | 7,0744 | 0,0237 | 0,0063 |

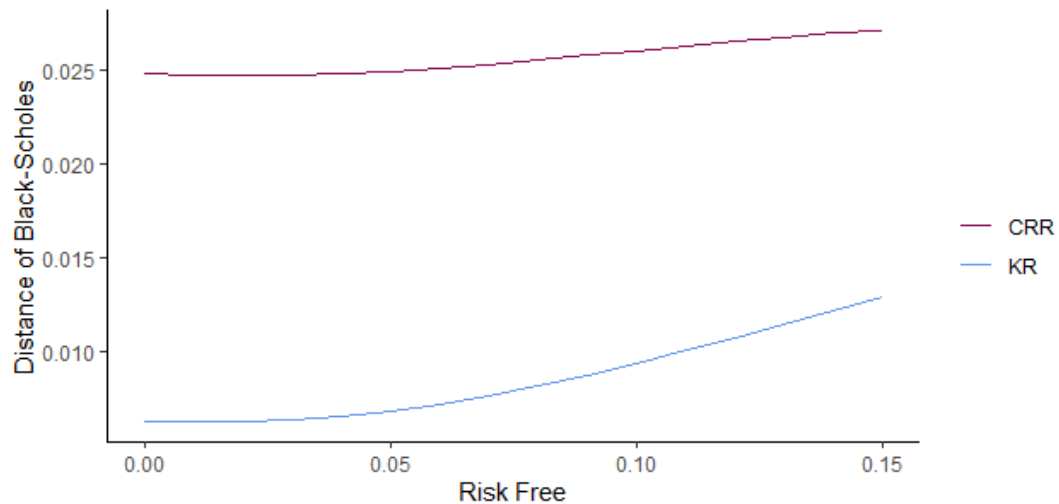


## RISK FREE ANALYSIS

| $S_0 = 100$ $K = 100$ $\text{Sigma} = 0.02$ $\text{Lamda} = 1.22474$ $\text{Time} = 1$ |        |        |         |        |
|----------------------------------------------------------------------------------------|--------|--------|---------|--------|
| Number of iterations                                                                   | CRR    | KR     | DiffCRR | DiffKR |
| Risk free = 0.01      BS = 7,4383                                                      |        |        |         |        |
| 5                                                                                      | 7,8377 | 7,3476 | -0,3994 | 0,0907 |
| 10                                                                                     | 7,2427 | 7,3901 | 0,1956  | 0,0482 |
| 20                                                                                     | 7,3398 | 7,4137 | 0,0985  | 0,0246 |

|                                          |        |        |         |        |
|------------------------------------------|--------|--------|---------|--------|
| 40                                       | 7,3889 | 7,4259 | 0,0494  | 0,0124 |
| 60                                       | 7,4053 | 7,4300 | 0,0330  | 0,0083 |
| 80                                       | 7,4135 | 7,4321 | 0,0248  | 0,0062 |
| 100                                      | 7,4185 | 7,4333 | 0,0198  | 0,0050 |
| <b>Risk free = 0.04      BS = 6,0040</b> |        |        |         |        |
| 5                                        | 6,3746 | 5,9073 | -0,3706 | 0,0967 |
| 10                                       | 5,8078 | 5,9530 | 0,1962  | 0,0510 |
| 20                                       | 5,9052 | 5,9780 | 0,0988  | 0,0260 |
| 40                                       | 5,9544 | 5,9909 | 0,0496  | 0,0131 |
| 60                                       | 5,9709 | 5,9952 | 0,0331  | 0,0088 |
| 80                                       | 5,9792 | 5,9974 | 0,0248  | 0,0066 |
| 100                                      | 5,9841 | 5,9987 | 0,0199  | 0,0053 |
| <b>Risk free = 0.07      BS = 4,7809</b> |        |        |         |        |
| 5                                        | 5,0984 | 4,6651 | -0,3175 | 0,1157 |
| 10                                       | 4,5807 | 4,7208 | 0,2001  | 0,0601 |
| 20                                       | 4,6801 | 4,7504 | 0,1008  | 0,0304 |
| 40                                       | 4,7303 | 4,7655 | 0,0506  | 0,0153 |
| 60                                       | 4,7471 | 4,7706 | 0,0338  | 0,0102 |
| 80                                       | 4,7555 | 4,7732 | 0,0253  | 0,0077 |
| 100                                      | 4,7606 | 4,7747 | 0,0203  | 0,0062 |

**Risk Free Analysis European Put**

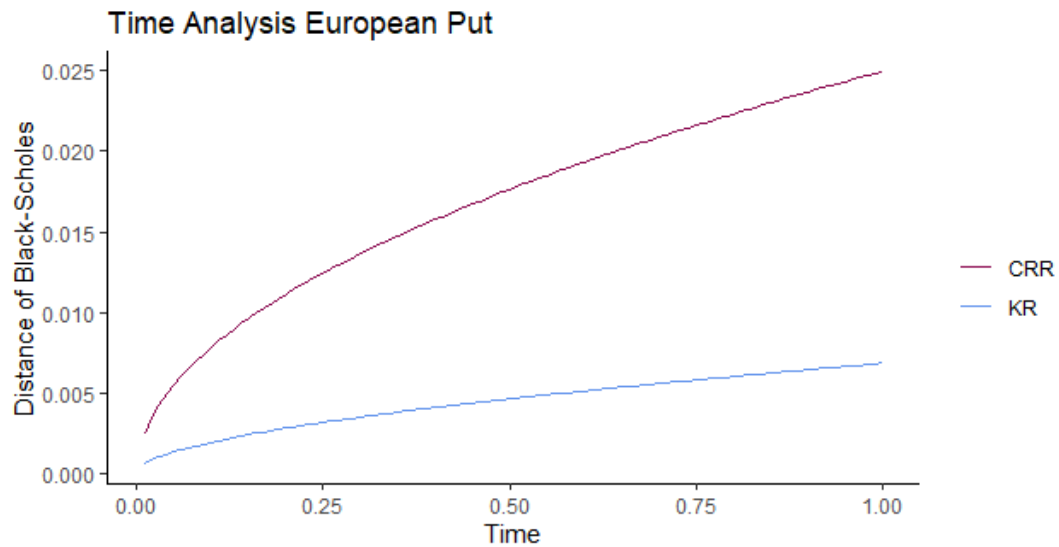


## TIME ANALYSIS

| $S_0 = 100$ $K = 100$ $\text{Sigma} = 0.02$ $\text{Lamda} = 1.22474$ $\text{Risk free} = 0.05$ |        |        |         |        |
|------------------------------------------------------------------------------------------------|--------|--------|---------|--------|
| Number of iterations                                                                           | CRR    | KR     | DiffCRR | DiffKR |
| <b>Time = 2/12      BS = 2,8449</b>                                                            |        |        |         |        |
| 5                                                                                              | 3,0072 | 2,8071 | -0,1623 | 0,0378 |
| 10                                                                                             | 2,7645 | 2,8248 | 0,0804  | 0,0200 |
| 20                                                                                             | 2,8044 | 2,8346 | 0,0405  | 0,0102 |
| 40                                                                                             | 2,8246 | 2,8397 | 0,0203  | 0,0052 |
| 60                                                                                             | 2,8313 | 2,8414 | 0,0136  | 0,0035 |
| 80                                                                                             | 2,8347 | 2,8423 | 0,0102  | 0,0026 |
| 100                                                                                            | 2,8367 | 2,8428 | 0,0081  | 0,0021 |
| <b>Time = 5/12      BS = 4,1289</b>                                                            |        |        |         |        |
| 5                                                                                              | 4,3771 | 4,0671 | -0,2482 | 0,0618 |
| 10                                                                                             | 4,0017 | 4,0962 | 0,1272  | 0,0327 |
| 20                                                                                             | 4,0648 | 4,1122 | 0,0641  | 0,0167 |
| 40                                                                                             | 4,0968 | 4,1205 | 0,0321  | 0,0084 |
| 60                                                                                             | 4,1074 | 4,1233 | 0,0214  | 0,0056 |
| 80                                                                                             | 4,1128 | 4,1247 | 0,0161  | 0,0042 |
| 100                                                                                            | 4,1160 | 4,1255 | 0,0129  | 0,0034 |
| <b>Time = 8/12      BS = 4,8942</b>                                                            |        |        |         |        |
| 5                                                                                              | 5,1978 | 4,8137 | -0,3036 | 0,0804 |
| 10                                                                                             | 4,7333 | 4,8518 | 0,1609  | 0,0424 |
| 20                                                                                             | 4,8131 | 4,8726 | 0,0811  | 0,0216 |

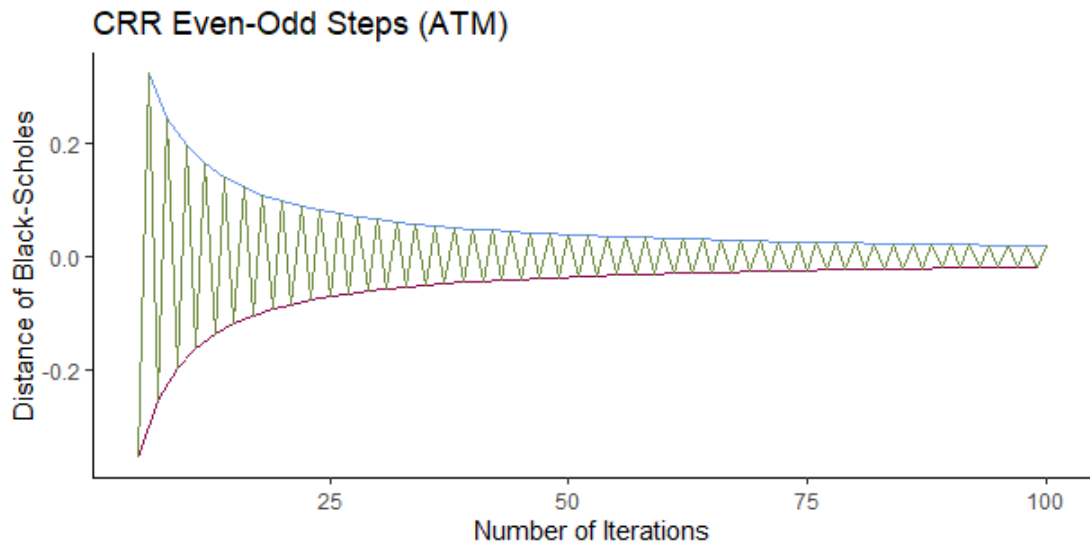


|     |        |        |        |        |
|-----|--------|--------|--------|--------|
| 40  | 4,8535 | 4,8833 | 0,0407 | 0,0109 |
| 60  | 4,8670 | 4,8869 | 0,0271 | 0,0073 |
| 80  | 4,8738 | 4,8887 | 0,0204 | 0,0055 |
| 100 | 4,8779 | 4,8898 | 0,0163 | 0,0044 |



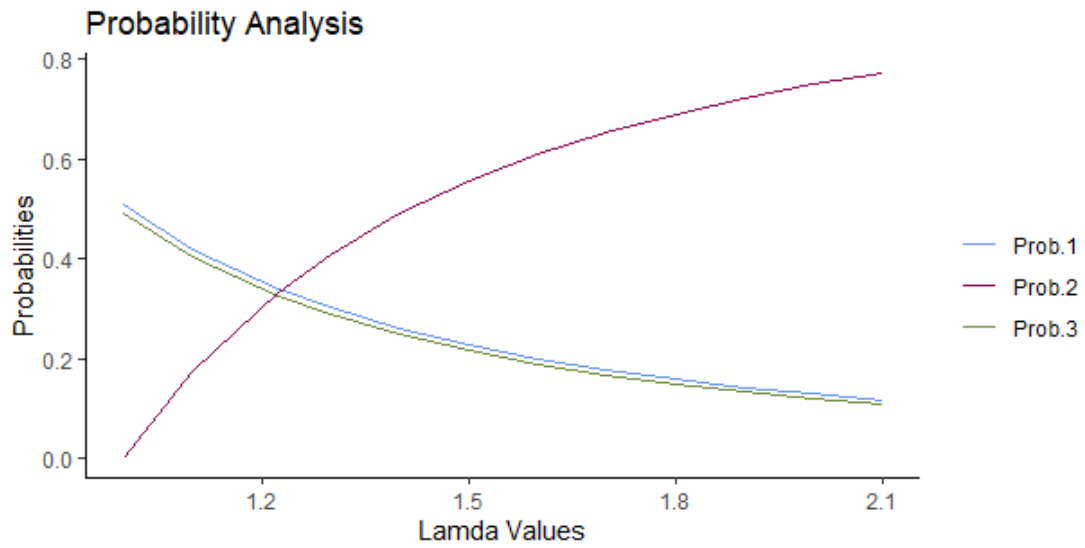
## CRR EVEN-ODD STEPS

## AT THE MONEY

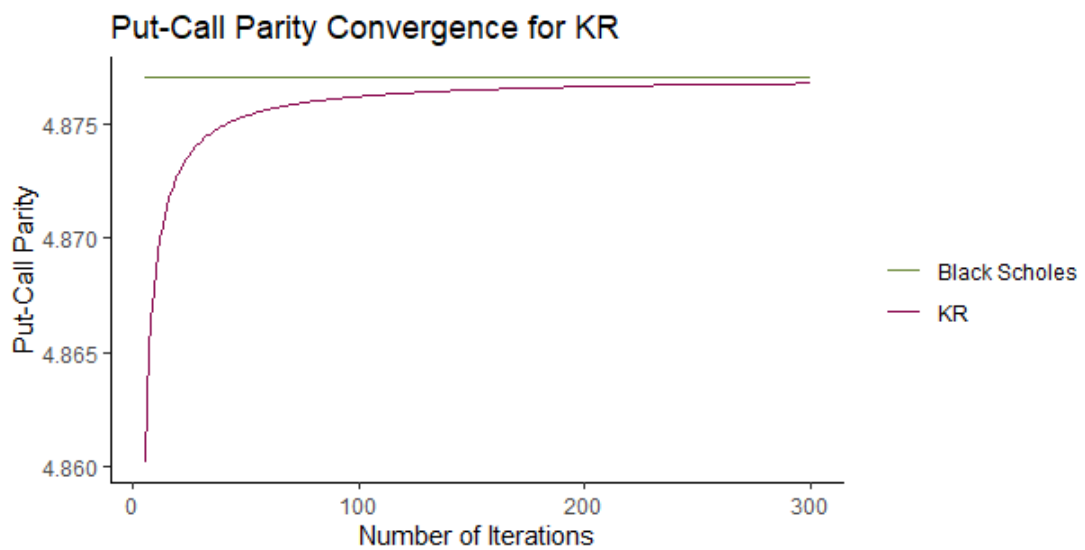


## **KR PROBABILITY ANALYSIS**

|        | 1       | 1.1     | 1.22474 | 1.4     | 1.7     | 1.9     | 2.1     |
|--------|---------|---------|---------|---------|---------|---------|---------|
| Prob.1 | 0,50839 | 0,42085 | 0,34018 | 0,26109 | 0,17794 | 0,14292 | 0,11737 |
| Prob.2 | 0,49161 | 0,40560 | 0,32648 | 0,24911 | 0,16808 | 0,13409 | 0,10939 |
| Prob.3 | 0,00000 | 0,17355 | 0,33332 | 0,48980 | 0,65398 | 0,72299 | 0,77324 |



## KR PUT-CALL PARITY CONVERGENCE



# **MINIMUM CONVERGENCE STEP FOR** **KR AND CRR**

**CALL**

| Call Option           |                  |                             |
|-----------------------|------------------|-----------------------------|
| So = 100              | Volatility = 0.2 | Time = 1 , Risk Free = 0.05 |
| Strike Price          | CRR              | KR                          |
| Precision level: 0.5% |                  |                             |
| 90                    | 16               | 4                           |
| 100                   | 39               | 12                          |
| 110                   | 71               | 18                          |
| Average               | 42.0             | 11.3                        |
| Precision level: 1%   |                  |                             |
| 90                    | 7                | 3                           |
| 100                   | 19               | 6                           |
| 110                   | 22               | 8                           |
| Average               | 16.0             | 5.6                         |
| Precision level: 5%   |                  |                             |
| 90                    | 2                | 1                           |
| 100                   | 4                | 1                           |
| 110                   | 6                | 3                           |
| Average               | 4.0              | 1.6                         |

## EUROPEAN PUT

| Put Option            |                  |                             |
|-----------------------|------------------|-----------------------------|
| So = 100              | Volatility = 0.2 | Time = 1 , Risk Free = 0.05 |
| Strike Price          | CRR              | KR                          |
| Precision level: 0.5% |                  |                             |
| 90                    | 99               | 40                          |
| 100                   | 71               | 20                          |
| 110                   | 40               | 8                           |
| Average               | 70.0             | 22.6                        |
| Precision level: 1%   |                  |                             |
| 90                    | 59               | 17                          |
| 100                   | 35               | 10                          |
| 110                   | 19               | 4                           |
| Average               | 37.6             | 10.3                        |
| Precision level: 5%   |                  |                             |
| 90                    | 7                | 4                           |
| 100                   | 7                | 3                           |
| 110                   | 3                | 2                           |
| Average               | 5.6              | 3.0                         |

## AMERICAN PUT

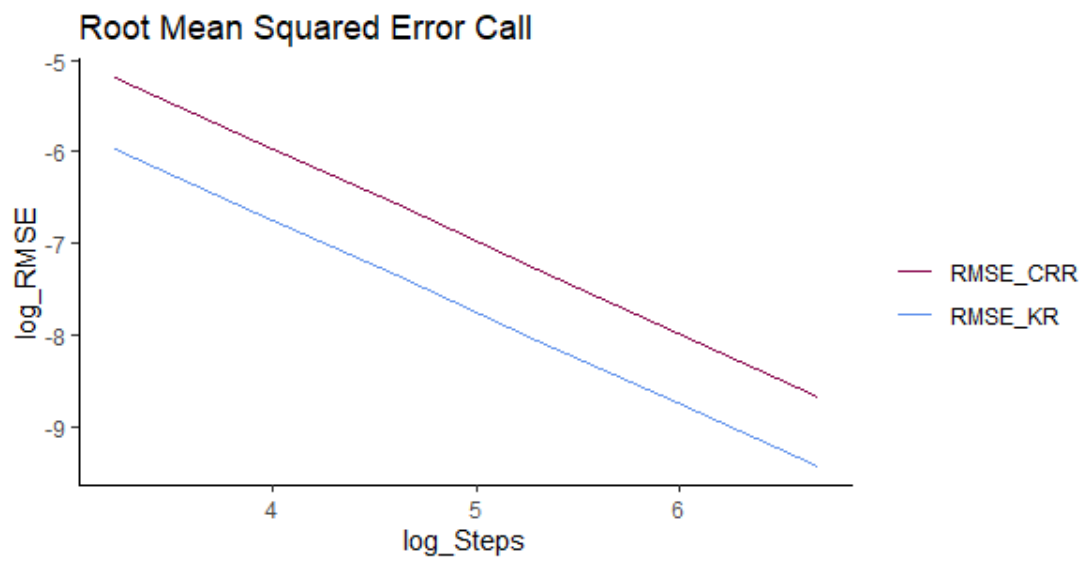
|                          |                        |                                |
|--------------------------|------------------------|--------------------------------|
| Volatility =<br>0.2      | American Put<br>Option | Time = 1 , Risk Free =<br>0.05 |
| <b>Strike<br/>Price</b>  | <b>CRR</b>             | <b>KR</b>                      |
| Precision level:<br>0.5% |                        |                                |
| <b>90</b>                | 70                     | 36                             |
| <b>100</b>               | 46                     | 13                             |
| <b>110</b>               | 16                     | 4                              |
| Average                  | 44.0                   | 17.6                           |
| Precision level: 1%      |                        |                                |
| <b>90</b>                | 41                     | 15                             |
| <b>100</b>               | 22                     | 6                              |
| <b>110</b>               | 5                      | 3                              |
| Average                  | 22.6                   | 8                              |
| Precision level: 5%      |                        |                                |
| <b>90</b>                | 7                      | 3                              |
| <b>100</b>               | 4                      | 2                              |
| <b>110</b>               | 2                      | 1                              |
| Average                  | 4.3                    | 2.0                            |

## Root Mean Squared Error

### Call Option

| Periods | RMSE_CRR   | RMSE_KR    |
|---------|------------|------------|
| 25      | 0,00550768 | 0,00257283 |
| 50      | 0,00275219 | 0,00128164 |
| 100     | 0,00139804 | 0,00064203 |
| 200     | 0,00068955 | 0,00031174 |

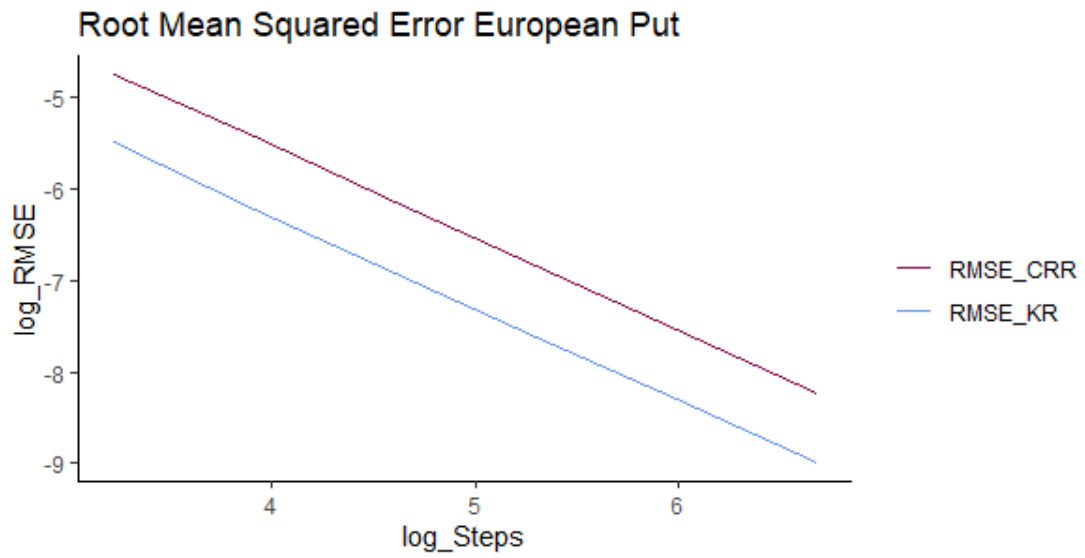
|     |            |            |
|-----|------------|------------|
| 400 | 0,00034254 | 0,00015898 |
| 800 | 0,00017079 | 0,00008018 |



## European Put Option

| Periods | RMSE_CRR   | RMSE_KR    |
|---------|------------|------------|
| 25      | 0,00848605 | 0,00413394 |
| 50      | 0,00438803 | 0,00196155 |
| 100     | 0,00215207 | 0,00097302 |
| 200     | 0,00106449 | 0,00049077 |
| 400     | 0,00053358 | 0,00025048 |
| 800     | 0,00026361 | 0,00012548 |





## American Put Option

| Periods | RMSE_CRR   | RMSE_KR    |
|---------|------------|------------|
| 25      | 0,00690142 | 0,00343888 |
| 50      | 0,00348670 | 0,00166291 |
| 100     | 0,00172567 | 0,00079874 |
| 200     | 0,00081608 | 0,00037697 |
| 400     | 0,00039078 | 0,00016888 |
| 800     | 0,00018873 | 0,00008021 |

